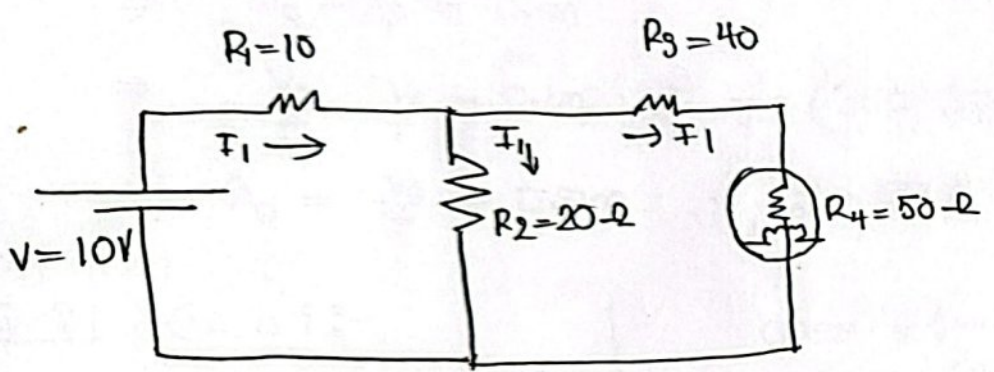


10/06/26
Lec-01

EEE101 - Belal Sir

Quiz: Probably 5 (n-1)

⇒ Make Up: Sir ko jawab karna hoga exm me,
Mid: Not the Main Mid Syllabus. Extra shakardar.



Topic: EMF, Pumping Charge, Voltage, Current

15/06/26
Lec-02

Potential diff
cause

Flow of Charge
Response

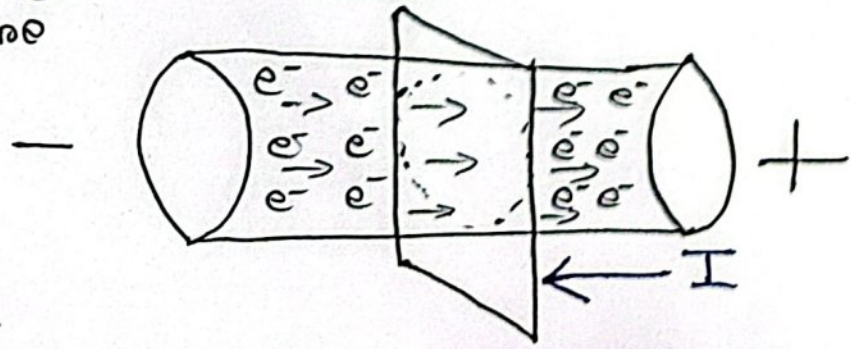
- Potential difference / EMF / Pressure (Cause)
- Flow of charge (Response)

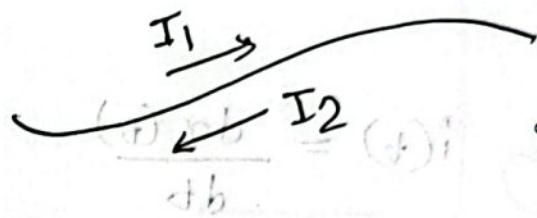
$$I = \frac{dq}{dt}$$

Ampere

→ +ve Charge flow

$1C \rightarrow 1s \rightarrow 1A$

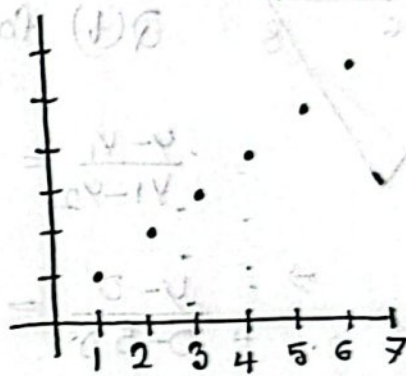




$$I_1 = -I_2$$

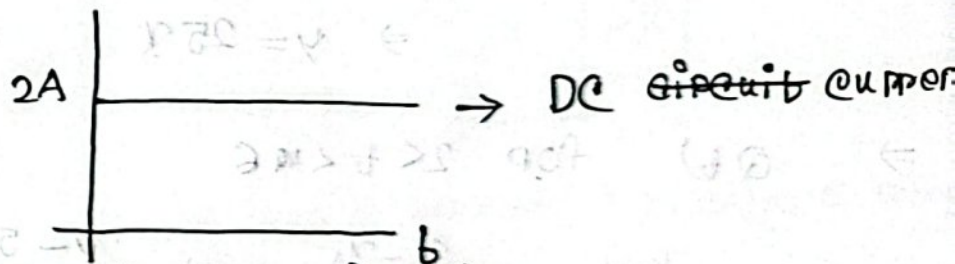
$$I_1 = 3A \quad I_2 = -3A$$

$$I(t) = \frac{dQ}{dt}(t)$$

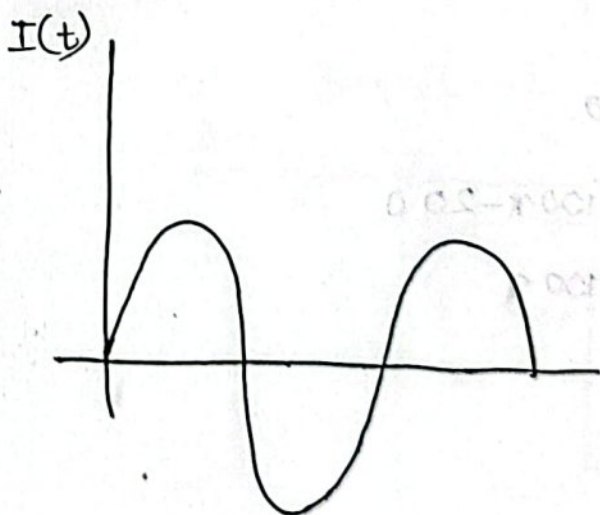


$$Q(t) = 2t$$

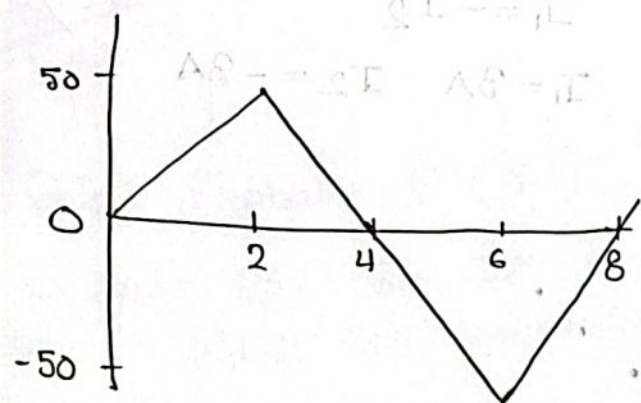
$$\therefore I = \frac{dQ(t)}{dt} = 2A$$



$\Rightarrow$ If current flow only in one direction.



alternating voltage



$$i(t) = \frac{dq(t)}{dt}$$

$Q(t)$ for $0 < t < 2$

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$y \rightarrow (Q(t))$$

 $x \rightarrow t$

$$\Rightarrow \frac{y - 0}{0 - 50} = \frac{x - 0}{0 - 2}$$

$$\Rightarrow \frac{y}{-50} = \frac{x}{-2} \Rightarrow 2y = 50x$$

$$\Rightarrow y = 25x$$

$\Rightarrow Q(t)$ for $2 < t < 4$

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2} \Rightarrow \frac{y - 50}{50 - (-50)} = \frac{x - 2}{2 - 4}$$

$$\Rightarrow \frac{y - 50}{100} = \frac{x - 2}{-2}$$

$$\Rightarrow -2y + 200 = 100x - 200$$

$$\Rightarrow -2y + 200 = 100x - 200$$

$$\Rightarrow -2y + 400 = 100x$$

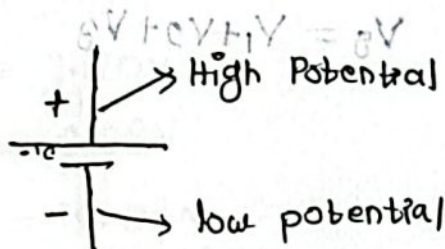
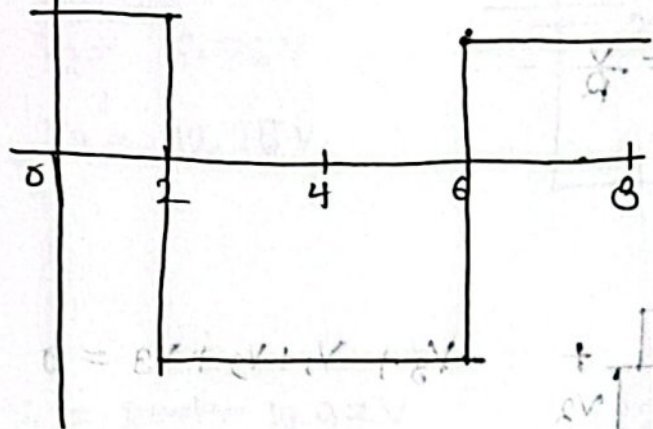
$$\Rightarrow \underline{\underline{-y + 200 = 50x}}$$

$$\Rightarrow -y + 200 = 50x$$

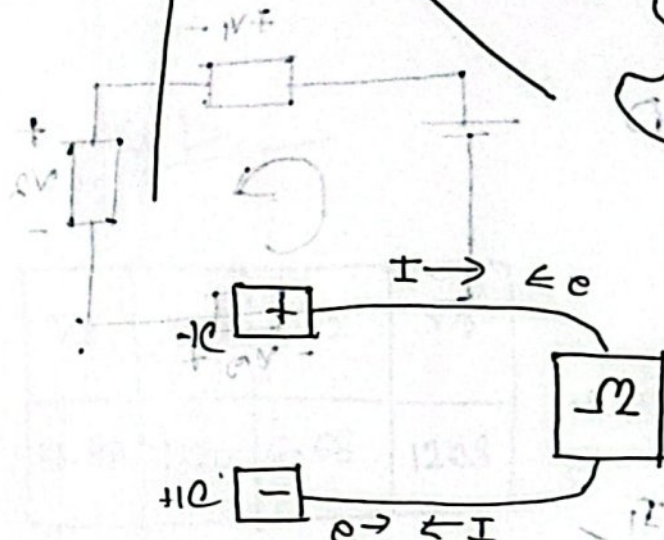
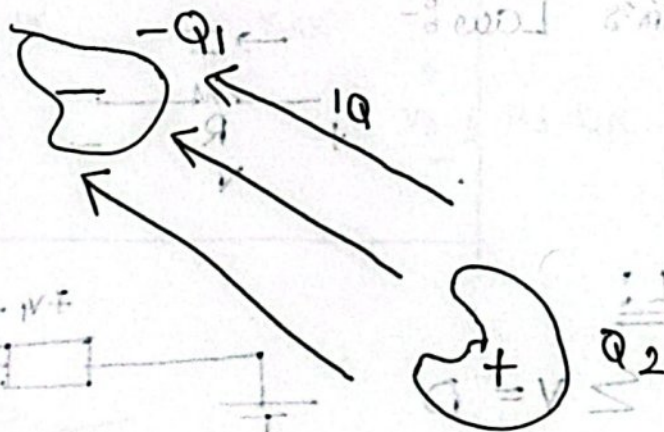
$$y - 200 = -50x$$

$$\Rightarrow y = -50x + 200$$

$I(t)$



$$F = -\frac{Q_1 Q_2}{4\pi \epsilon_0 r^2}$$



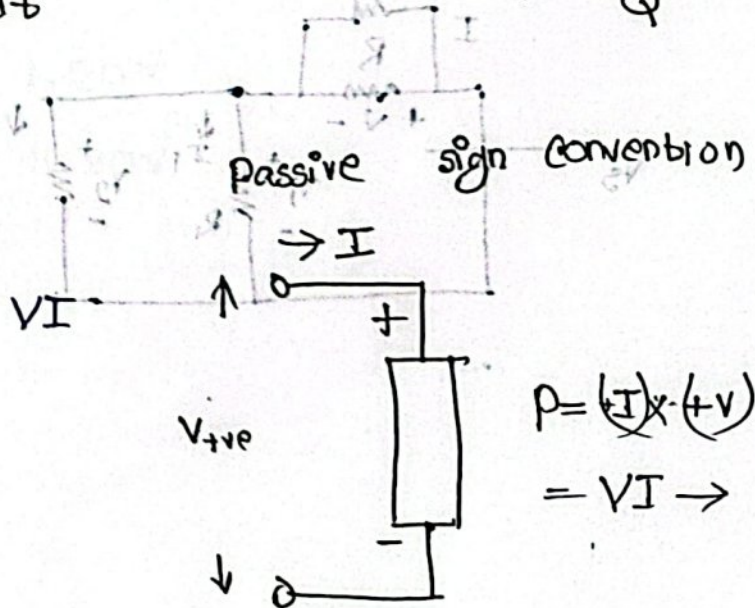
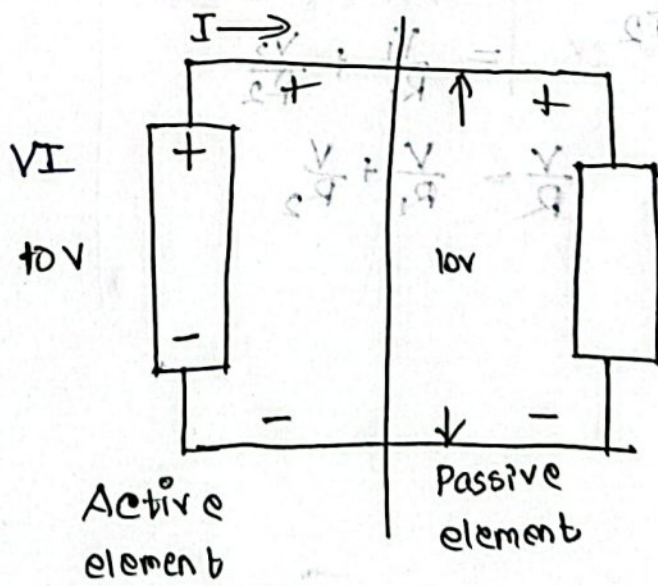
* Active element : supplies e energy

* Passive element : absorbs e "

Power : $p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = v i$

$$V = \frac{F}{Q}$$

$$P = VI = I^2 R$$



$$P = (I) \times (+V) = VI \rightarrow$$

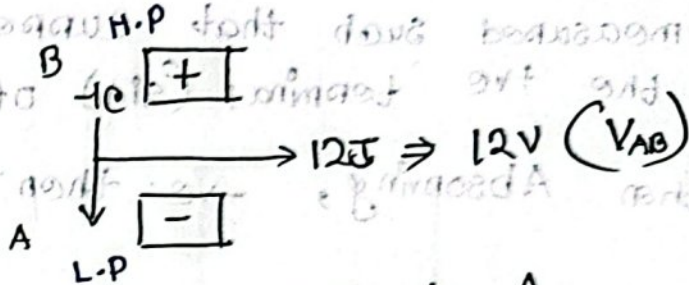
Lec-03
17/08/26

EEE101

$$I(t) = \frac{dQ(t)}{dt}$$

Cause
Voltage diff

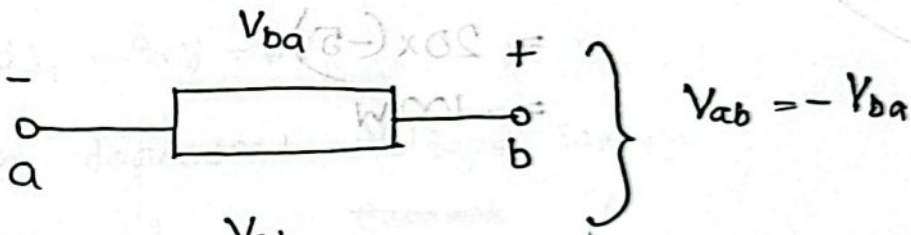
$$I_A = \frac{110}{15}$$



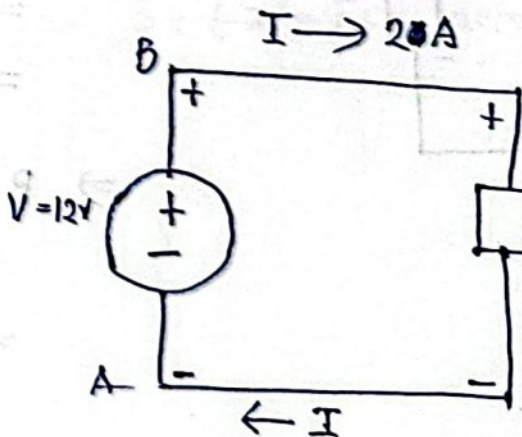
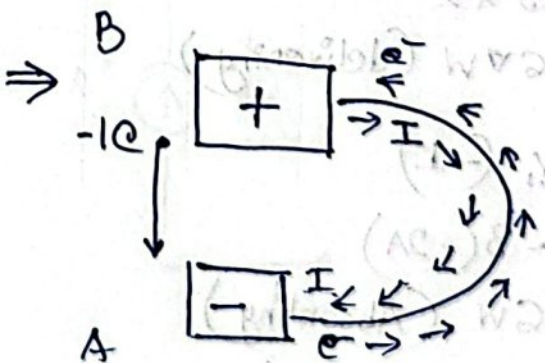
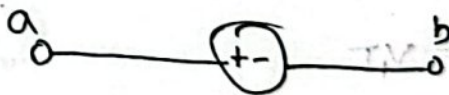
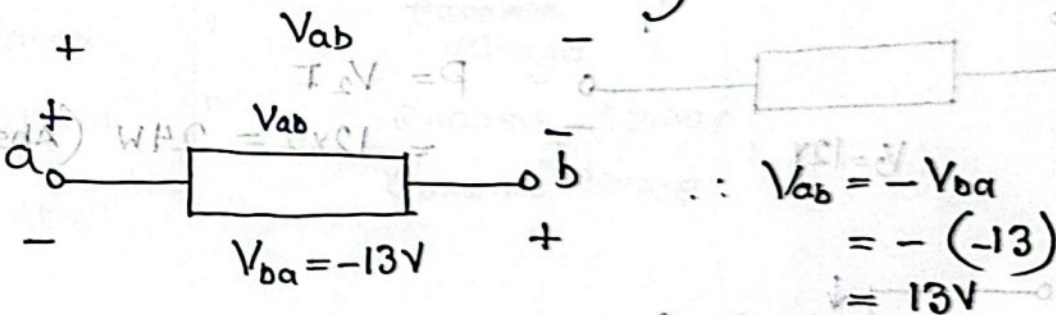
$V_{BA} \Rightarrow$ B's Voltage with respect to A.

$$V_{BA} = -V_{AB}$$

Q1



Q1



$$P = VI$$

$$= 12 \times 2$$

$$= 24W$$

$$P_2 = VI$$

$$= 12 \times 2$$

$$= 24W$$

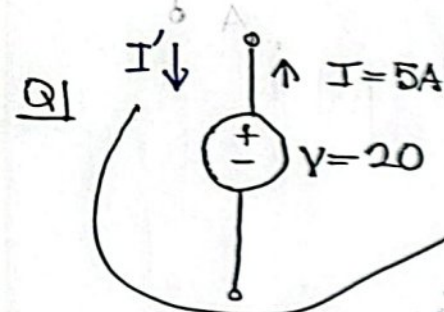
* Passive Sign Convention

Absorbing: $VI \rightarrow +ve$

Delivering: $VI \rightarrow -ve$

$\Rightarrow$ Power is measured such that current should always flow into the +ve terminal (side) of the element.

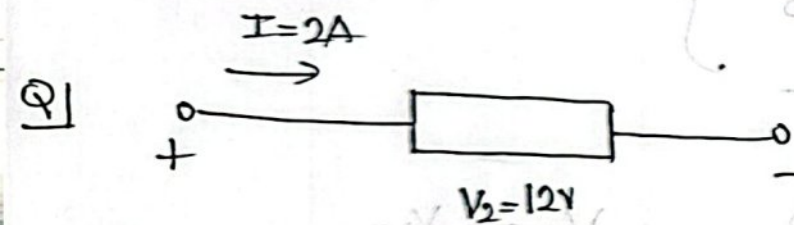
If +ve then Absorbing, -ve then delivering.



$$P = VI' = V(-I)$$

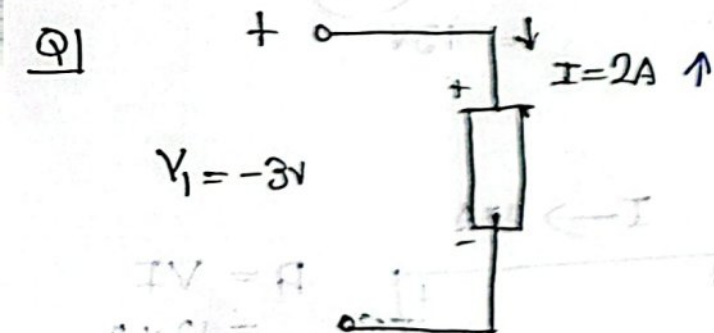
$$= 20 \times (-5)$$

$$= -100W$$



$$P = V_2 I$$

$$= 12 \times 2 = 24W \text{ (Absorbing)}$$



$$\Rightarrow P = V_1 I$$

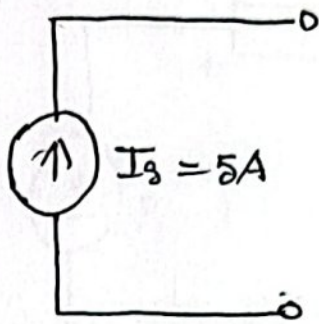
$$= -3 \times 2$$

$$= -6W \text{ (delivering)}$$

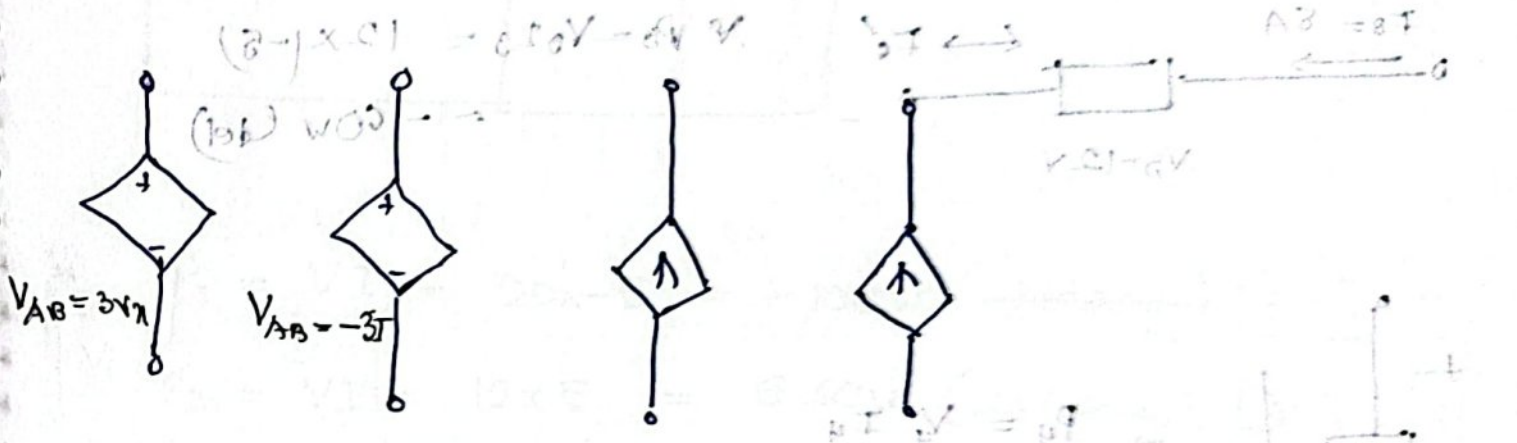
$$\Rightarrow P = V_1 (-I)$$

$$= -3(-2A)$$

$$= 6W \text{ (Absorbing)}$$

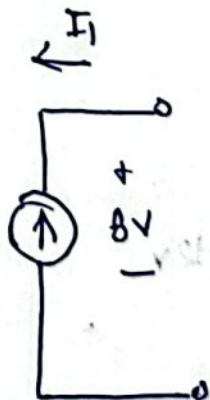
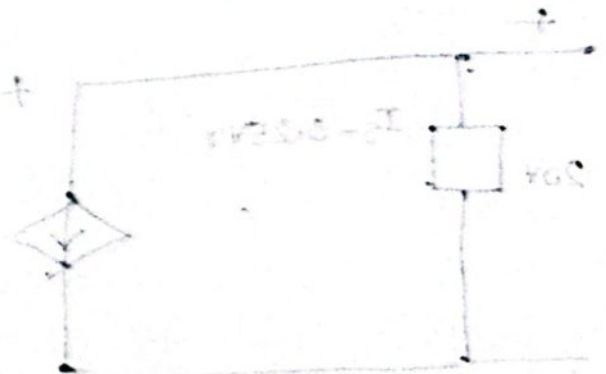


providing constant current regardless the voltage across it.



$$V_1 = 3I_1 = 3 \times 8 = 24V$$

- ⇒ Voltage dependent Voltage source
- ⇒ Current " ~~Current~~ Voltage "
- ⇒ Current " Current source
- ⇒ voltage " Current source



$$P = V_1 I_1 = 8 \times (-7) = 56W \text{ (delivered)}$$

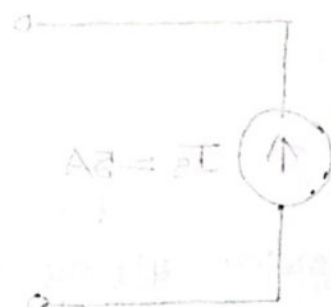
$$I_2 = 2A$$



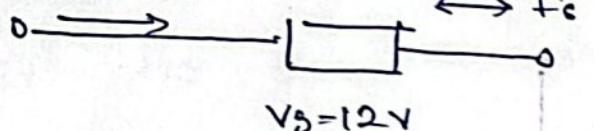
$$v_2 = 8V$$

$$\Rightarrow P_2 = 12W$$

$$= 12V \times 1A = 12W [Abs]$$



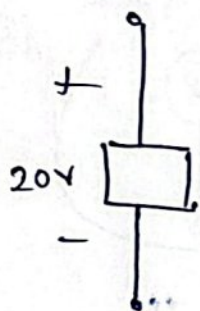
$$I_3 = 5A$$



$$v_3 = 12V$$

$$P_3 = V_3 I_3 = 12 \times (-5)$$

$$= -60W (del)$$

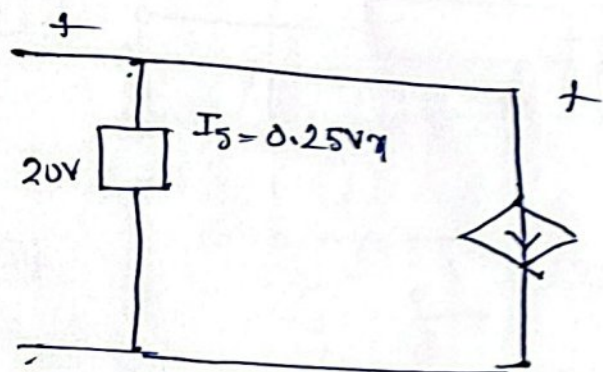


$$\Rightarrow I_4 = 8A$$

$$P_4 = V_4 I_4$$

$$= 160W (absorbing)$$

$$V_4 = 20V, I_4 = 8A$$



$$I_5 = 0.25A$$

$$20V$$

$$P_5 = V_5 I_5$$

$$= 20 \times 0.25A$$

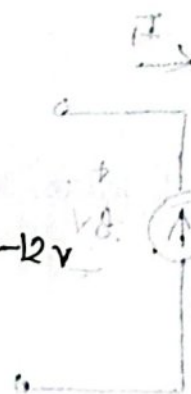
$$V_p = -V_x \quad 20V$$

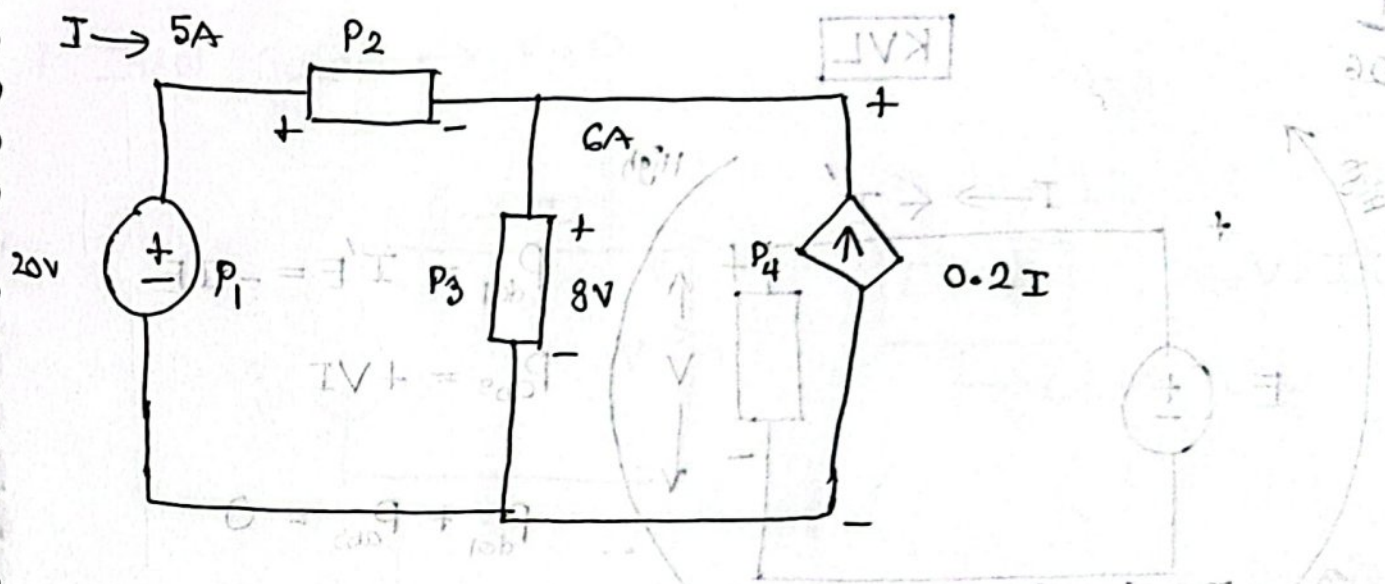
$$V_p = V_3 = 12V$$

$$\leftarrow V_x = 12V \quad \text{or} \quad V_x = -12V$$

$$P_5 = 20 \times 0.25 \times (-12)$$

$$= -60W (del)$$





$$P_1 = VI = 20 \times 5 = -100 \text{ W (Absorbing) [del] [del]}$$

$$P_2 = VI = 12 \times 5 = 60 \text{ W (delivering) [Ab]}$$

$$P_3 = VI = 6 \times 8 = 48 \text{ W (delivering) [Ab]}$$

$$P_4 = VI = -1 \times 8 = -8 \text{ W (Absorbing) [del] [del]}$$

Flow of Charge (Current) → Voltage / Potential diff. (Response)



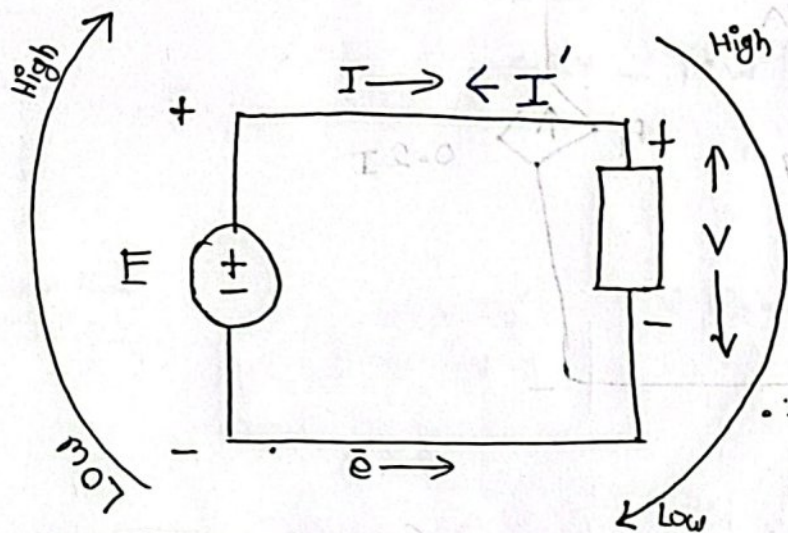
$$I = \frac{V}{R} \Rightarrow \frac{V}{R} = I$$



$$R = \frac{V}{I}$$

Lec-04
22/06/26

KVL



$$P_{del} = I'E = -IE$$

$$P_{abs} = +VI$$

$$P_{del} + P_{abs} = 0$$

$$\Rightarrow -EI + VI = 0$$

$$\Rightarrow -E + V = 0 \quad \therefore V = E$$

$$+E - V = 0$$

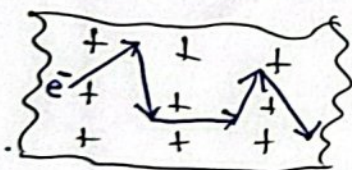
Rise in Potential Vol

Drop in Potential Voltage

Low to High $\longrightarrow$ Rise in Potential

High to Low $\longrightarrow$ Drop "

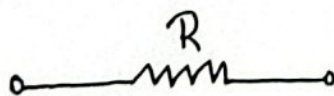
Voltage/Potential diff $\longrightarrow$ Flow of Charge (Current)
(Cause) (Response)



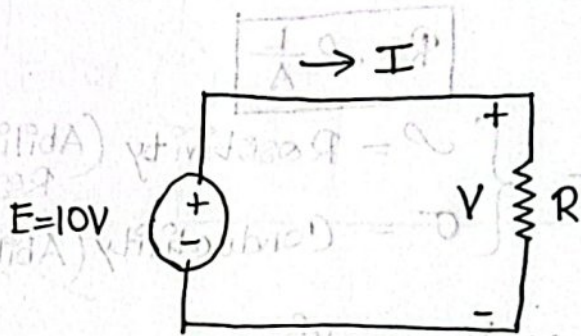
* e^- Movement is never straight forward $\longrightarrow$ it collides

$$I = \frac{\text{Cause (P.D)}}{\text{Opposition}} \Rightarrow I = \frac{E}{R} \quad \text{Resistance } (\Omega)$$

$$R = \frac{E \rightarrow 1V}{I \rightarrow 1A} \} 1\Omega$$



* Ideal Case $\rightarrow R=0$

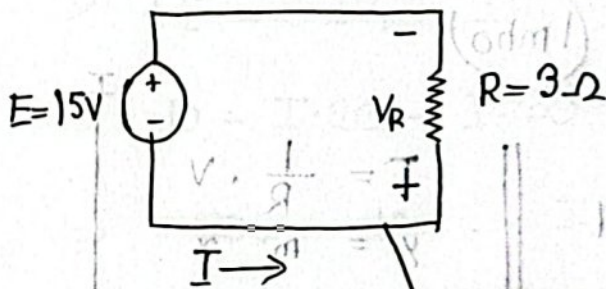


$$E = V = IR$$

$$I = \frac{E}{R} \quad \therefore E = V = IR$$

$\rightarrow$ Ohm's Law

* Current Enters in a Passive element via Positive terminal



* Summation of Voltage around circuit (complete) = 0

* Low to High $\rightarrow$ +ve

* High to Low $\rightarrow$ -ve

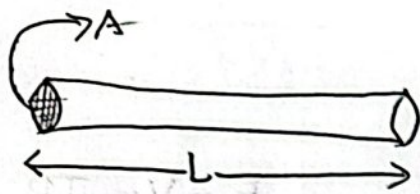
$$\therefore +E + V_R = 0 \quad [KVL]$$

$$\Rightarrow 15 + V_R = 0$$

$$\Rightarrow 15 + IR = 0 \quad [Apply \text{ Ohm's Law}]$$

$$\Rightarrow 15 + 3I = 0$$

$$\Rightarrow I = -5A$$



$$R \propto \frac{L}{A} \begin{bmatrix} R \propto L \\ R \propto \frac{1}{A} \end{bmatrix}$$

$$R = \rho \frac{L}{A}$$

ρ = Resistivity (Ability to Resist)
 σ = Conductivity (Ability...)

$$P = VI = I \cdot R \cdot I = I^2 R \quad \Bigg| = V \cdot \frac{V}{R} = \frac{V^2}{R}$$

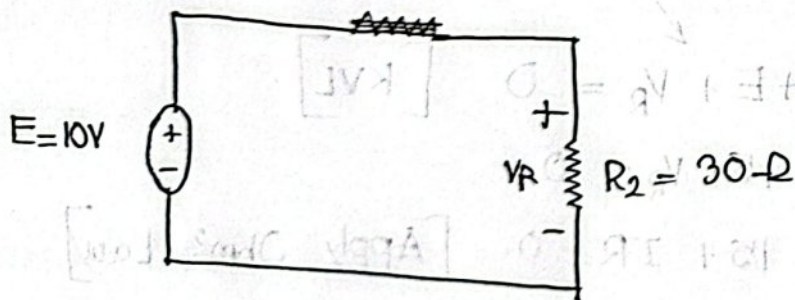
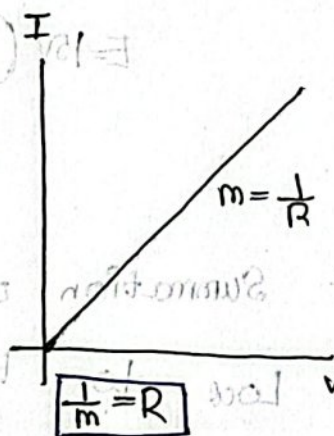
$$I = \frac{V}{R} \Rightarrow R = \frac{V}{I} \therefore \frac{I}{V} = \frac{1}{R} = G \text{ (Conductance)}$$

$$1s = \frac{1A}{1V} \therefore 1s = 1 \Omega \cdot (1mho)$$

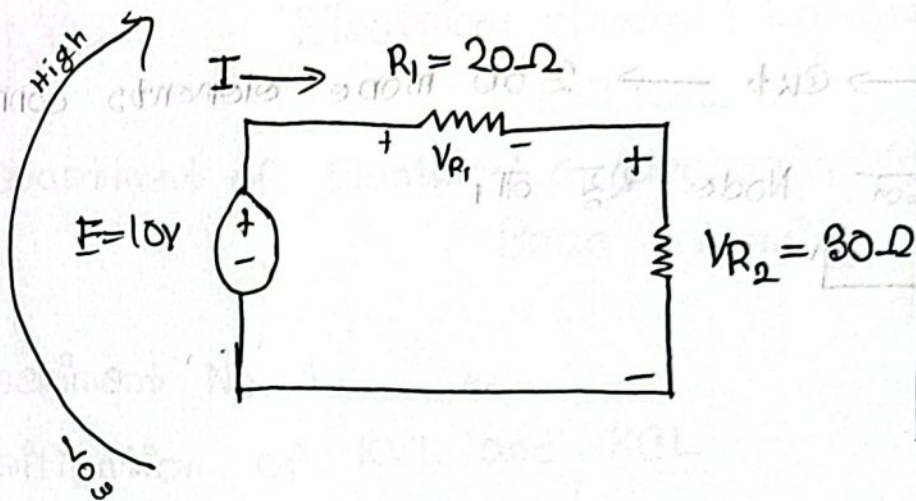
$$\therefore P = VI = \frac{I^2}{GR} = V^2 G$$

$$I = \frac{1}{R} \cdot V$$

$$y = m x$$



$$\Rightarrow I = \frac{V}{R} = \frac{10}{30}$$



- * Assign current I
- * Polarity Set
- * KVL
- * Ohm Law

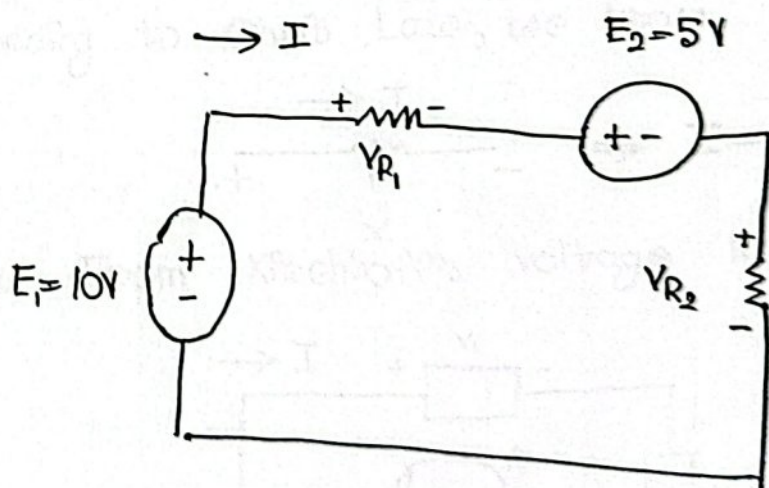
* Same current $\longrightarrow$ connected through series.

$$+E - V_{R_1} - V_{R_2} = 0 \quad [\text{KVL}]$$

$$\Rightarrow +E - IR_1 - IR_2 = 0 \quad [\text{apply ohm's law}]$$

$$\Rightarrow 10 - I \cdot 20 - I \cdot 30 = 0$$

$$\Rightarrow -50I = -10 \quad \therefore I = \frac{10}{50} = 0.2 \text{ A}$$



$$\Rightarrow +E_1 - V_{R_1} - E_2 - V_{R_2} = 0$$

$$\Rightarrow 10 - 20I - 5 - 30I = 0$$

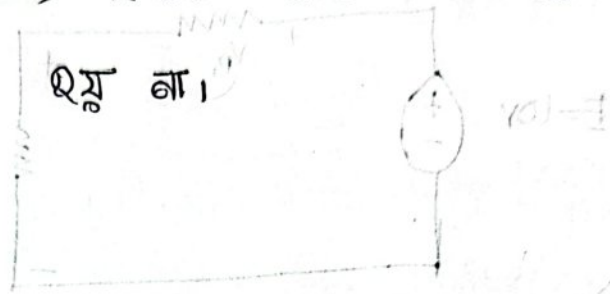
$$\Rightarrow 5 - 50I = 0$$

$$\therefore I = \frac{-5}{-50} = 0.1 \text{ A}$$

* Node : A point $\rightarrow$ out $\rightarrow$ 2 or more elements connected

* Closed CKT का दोन Node हय गा।

* Floating Node $\rightarrow$



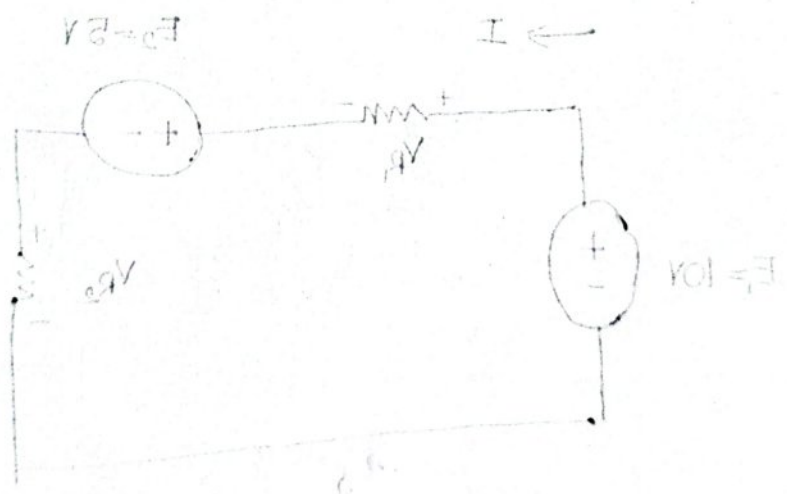
* Some Concept $\rightarrow$ Connected through series

$$[KVL] \quad 0 = V_1 - V_2 - E +$$

$$\Rightarrow +E - IR_1 - IR_2 = 0 \quad [apply KVL law]$$

$$\Rightarrow 10 - I \cdot 20 - I \cdot 20 = 0$$

$$\Rightarrow -20I = -10 \quad I = \frac{10}{20} = 0.5A$$



$$\Rightarrow +E_1 - V_1 - I_2 - V_2 = 0$$

$$\Rightarrow 10 - 20I - 2 - 20I = 0$$

$$\Rightarrow 2 - 20I = 0$$

$$I = \frac{2}{20} = 0.1A$$

Lee-05
24/06/26

EEE101

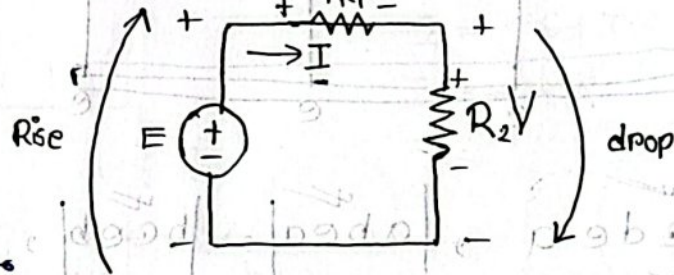
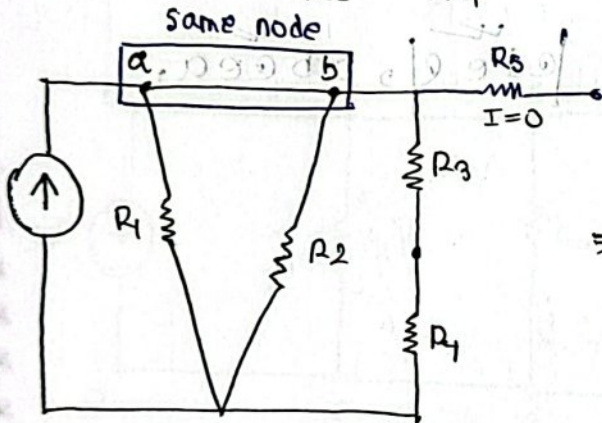
BEL

$$I = \frac{\text{Cause}}{\text{Opposition}} = \frac{V}{R}$$

$$[1.2 = \frac{1V}{1A}, \text{Ohm's Law}]$$

$$-EI = VI \Rightarrow \uparrow +E - V \downarrow = 0$$

Rise Drop



$$\Rightarrow +E - VR_1 - VR_2 = 0 \quad [KVL]$$

$$\Rightarrow E - IR_1 - IR_2 = 0$$

$$\Rightarrow I = \frac{E}{R_1 + R_2}$$

* No matter How many elements are between, if they are ~~same~~ in straight manner/path they are same node.

* Same Node, Same Voltage.

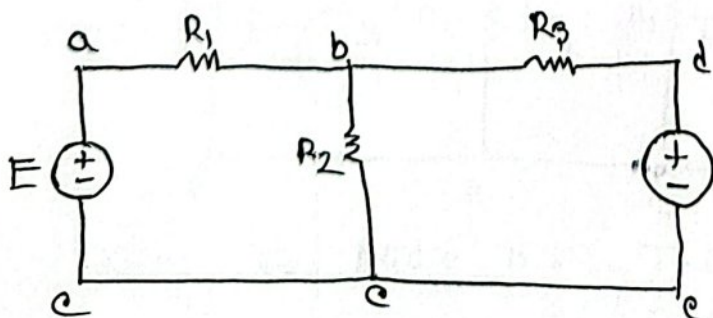
* $-R_5 \Rightarrow$ Floating Node

* Branch: A path through which same current flows.

* In a Branch All the elements are connected in series

* Same I — series || Same V — Parallel

* Closed Path = loop



$$a \rightarrow b \rightarrow c \rightarrow a \quad [\text{loop}]$$

loop 1: $[a] b c [a]$ [Independent]

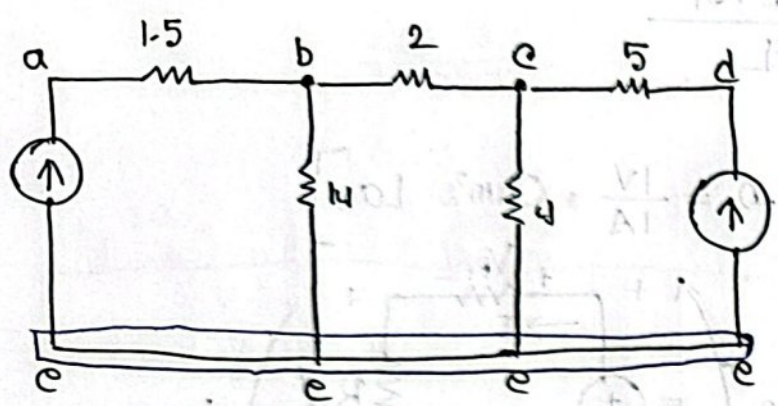
loop 2: $[a] b d c [a]$

loop 3: $[b] d c [b]$ [Independent]

encountered more than once

10-02-2020

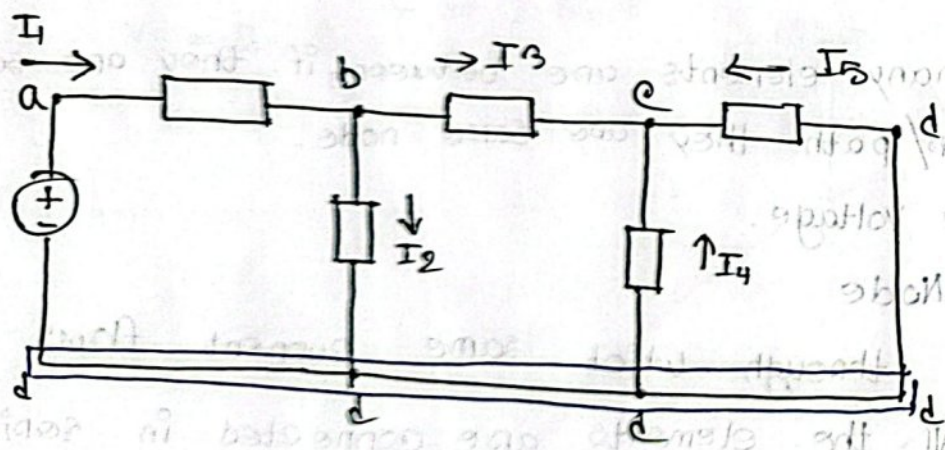
Q1



5 Nodes
Branch [Same Current]

$\Rightarrow$ $abcedea$, $abcea$, $bceeb$, $cedee$, $abcea$, $bcedeb$

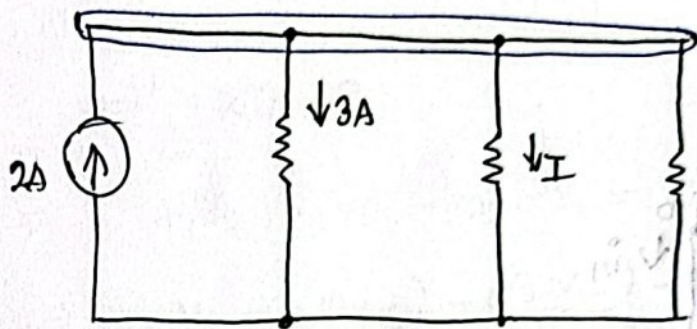
Topic: KCL



Node b: $I_1 - I_2 - I_3 = 0 \Rightarrow I_1 = I_2 + I_3$ (in = out)

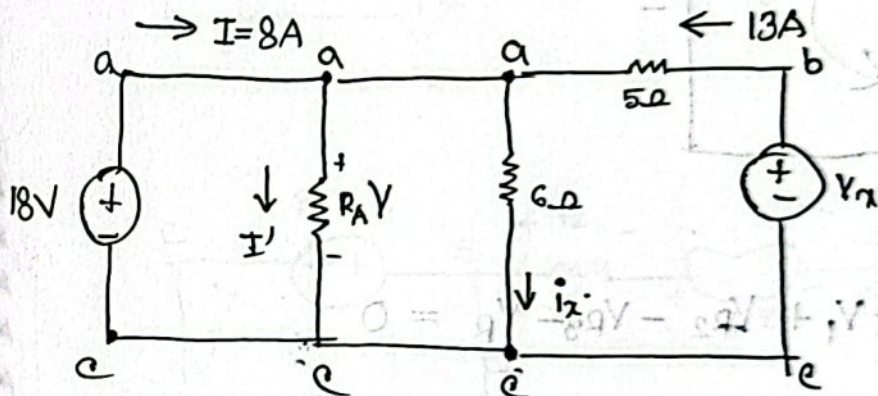
Node c: $I_3 + I_4 + I_5 = 0$





$$\begin{aligned} \text{I.C.} &= 2A \\ \text{O.C.} &= 3 + I + 3 \quad [\text{KCL}] \end{aligned}$$

$$\begin{aligned} 2 &= 6 + I \\ I &= -4A \end{aligned}$$



$$\begin{aligned} \text{KCL @ Node } a: \quad \text{I.C.} &= 8 + 3 = 21A \\ \text{O.C.} &= I' + 3A \end{aligned} \quad \left. \begin{aligned} 21 &= I' = 3 \\ I' &= 18A \end{aligned} \right\}$$

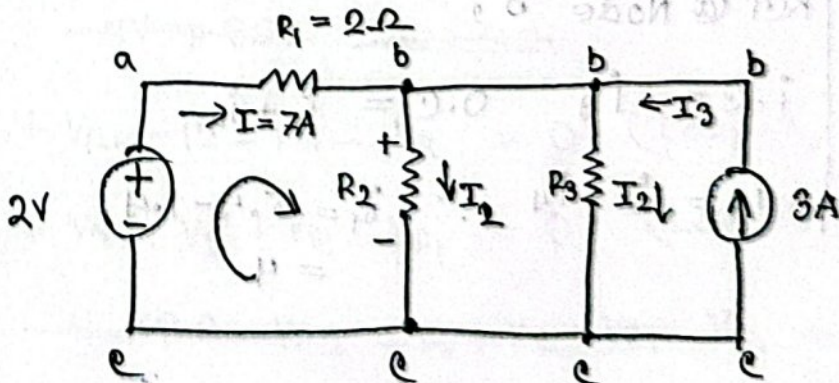
$$\text{KVL @ Node } c: +18 - V_{R_A} = 0 \quad [\text{KVL}]$$

$$+18 - I' R_A = 0$$

$$\Rightarrow 18 - 18 R_A = 0$$

$$R_A = 1\Omega$$

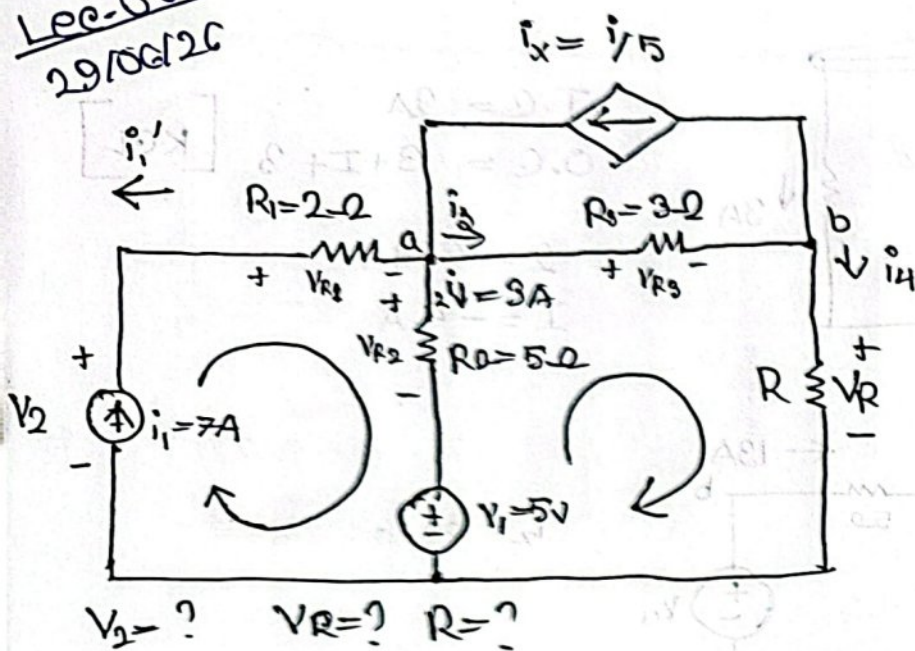
Q1



$$\begin{aligned} \text{KCL @ Node } b: \quad \text{I.C.} &= 7A + 3A \\ \text{O.C.} &= 7 \end{aligned}$$

Quiz: Next Wednesday

Lec-06
29/06/20



$$i_x = \frac{i_1}{5} = \frac{7}{5} = 1.4A$$

KCL @ Node a,

$$i.e.: i_1 + i_x$$

$$O.C.: i_2 + i_3$$

$$i.e. = 0$$

$$\Rightarrow i_1 + i_x = i_2 + i_3$$

$$\Rightarrow 7 + 1.4 = 3 + i_3$$

$$i_3 = 5.4A$$

$$+V_1 + V_{R2} - V_{R3} - V_R = 0$$

$$\Rightarrow 5 + i_2 R_2 - i_3 R_3 - V_R = 0$$

$$\Rightarrow \boxed{V_R = 3.8V}$$

$$R = \frac{V_R}{i_4} = \frac{3.8}{4} = \frac{3.8}{4} = \boxed{0.95\Omega}$$

KCL @ Node b,

$$i.e. = i_3 \quad O.C. = i_x + i_4$$

$$i_3 = i_x + i_4 \quad \therefore i_4 = 5.4 - 1.4 = 4$$

* Betw 2 Nodes in a circuit there must be a voltage.

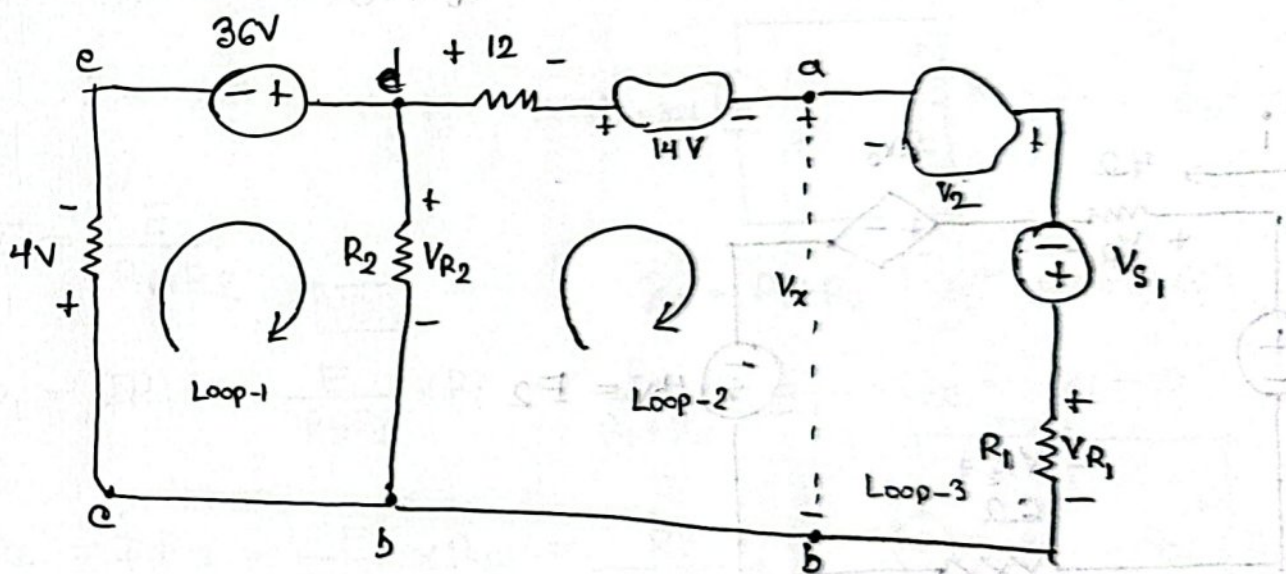
~~KVL~~ @

$$\Rightarrow +V_2 - V_{R_1} - V_{R_2} - V_1 = 0$$

$$\Rightarrow V_2 - (2 \times 7) - (3 \times 5) - 5 = 0$$

$$\boxed{V_2 = 34 \text{ V}}$$

$$P = V_2 i_1' = 34(-7) = -238 \text{ W (delivered)}$$



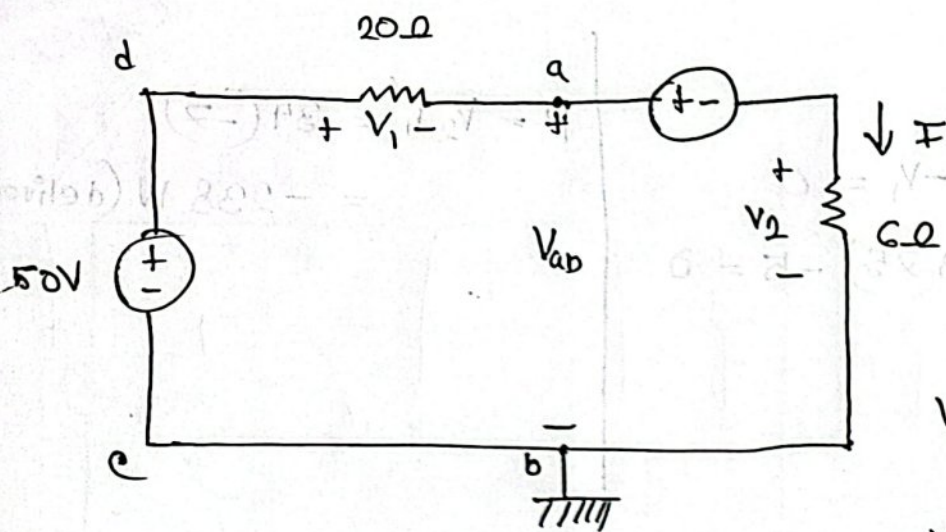
KVL @ Loop-2

$$\Rightarrow \cancel{4V + 36V} - \cancel{4V + 36V}$$

$$+V_{R_2} - 12 - 14 - V_x = 0 \quad (\text{Loop-2})$$

$$+V_x + V_2 + V_{s1} - V_{R_1} = 0 \quad (\text{Loop-3})$$

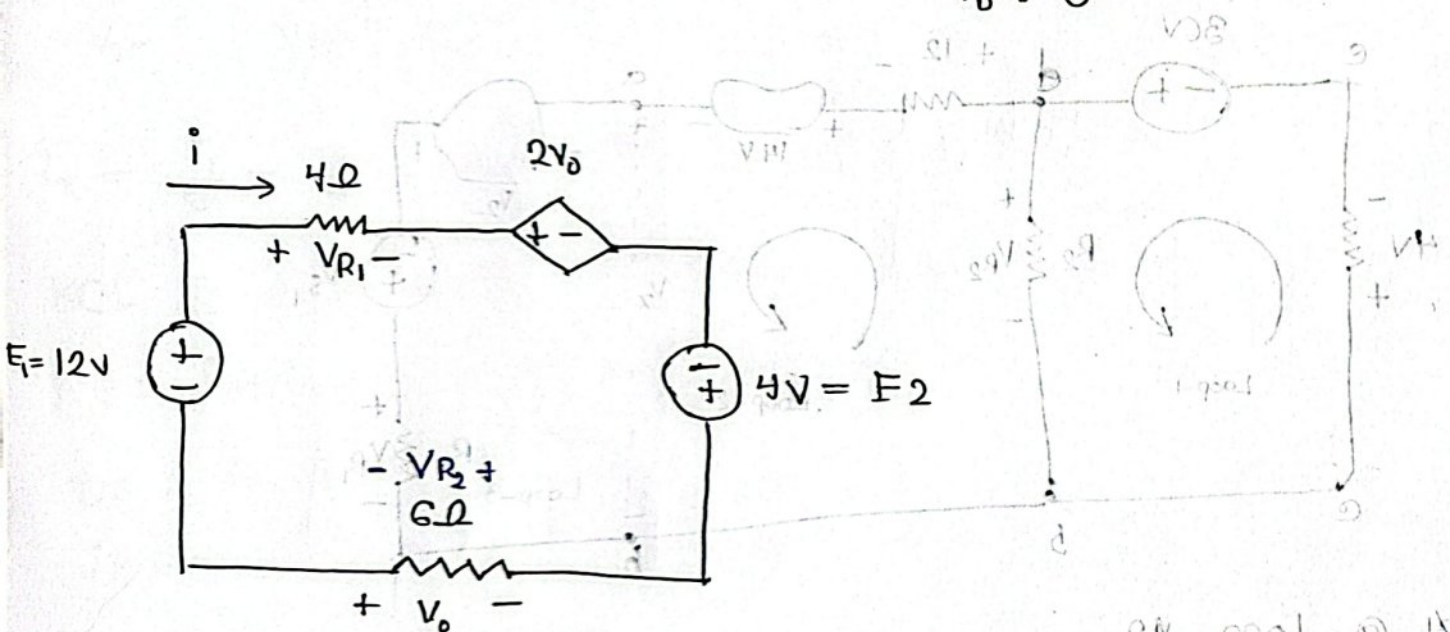
$$\cancel{4 + 36} - V_{R_2} + V_x + V_2 + V_{s1} - V_{R_1} = 0$$



$$V_{ab} = V_a - V_b$$

V_a : Voltage with respect to ground

$$V_b = 0$$



$$+ E_1 - V_{R1} - 2V_0 + E_2 - V_{R2} = 0$$

$$\Rightarrow 12 - 4I - 2V_0 + 4 - 6I = 0$$

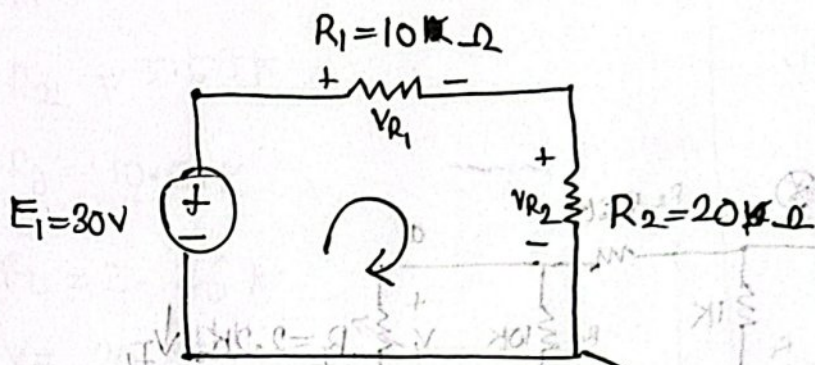
$$\Rightarrow 10I + 2V_0 = 16 \quad (1)$$

$$\Rightarrow 10 + 2(-6I) = 16$$

$$\Rightarrow 10I - 12I = 16$$

$$\Rightarrow I = -8A$$

$$V_0 = -V_{R2}$$



$$\nabla E_1 - V_{R1} - V_{R2} = 0$$

$$\Rightarrow 30 - 10I - 20I = 0$$

$$I = 1A$$

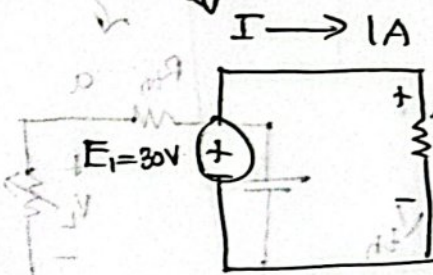
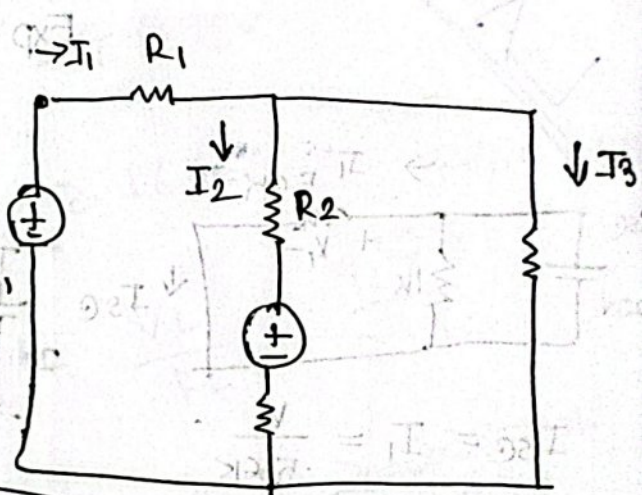
$$I = \frac{E}{R_1 + R_2}$$

[VDR]

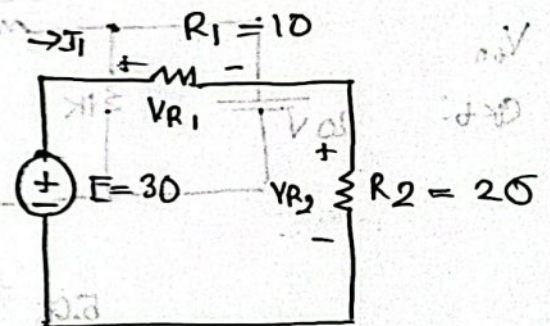
$$V_{R1} = IR_1 = \frac{E}{R_1 + R_2} \times R_1 = \frac{R_1}{R_1 + R_2} \times E$$

$$V_{R2} = IR_2 = \frac{E}{R_1 + R_2} \times R_2 = \frac{R_2}{R_1 + R_2} \times E$$

$$V_{R3} = IR_3 = \frac{E}{R_1 + R_2 + R_3} \times R_3$$



$$\therefore R_3 = R_1 + R_2 \text{ [eq. resistance]}$$

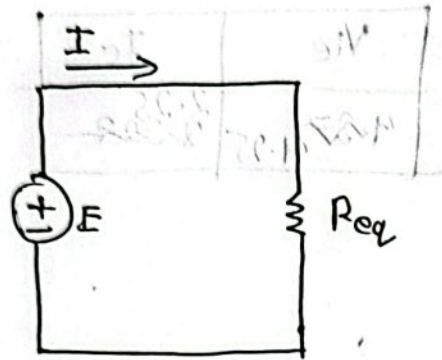


$$\therefore V_{R1} = 10V$$

$$V_{R2} = 20V$$

Quiz: Power, Circuit, KVL, KCL etc, sketch,

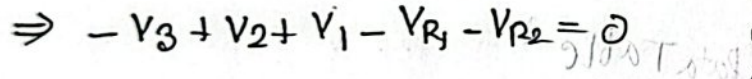
Lea-07



$$19 = 3.5 \times$$

100.0 = 100

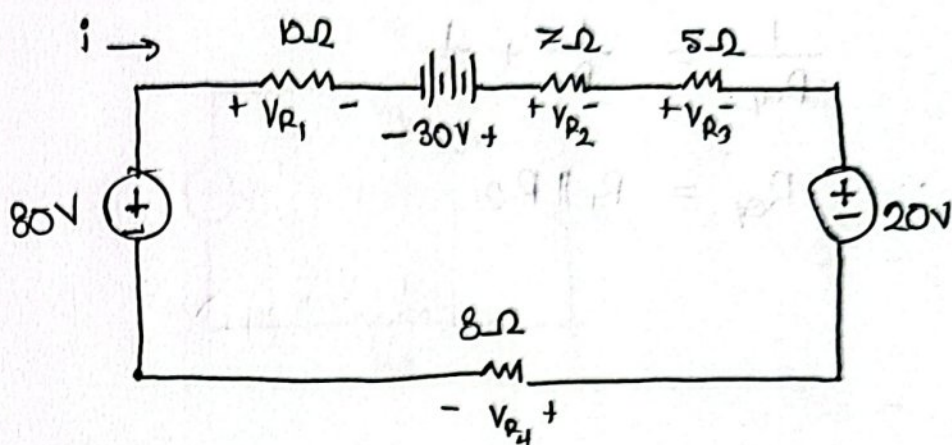
$$g \circ \phi = \mu g$$



$$\Rightarrow -10 + 9 + 15 - IR_1 - IR_2 = 0$$

$$\Rightarrow 14 - IR_1 - IR_2 = 0 \quad (2)$$

[illegible]

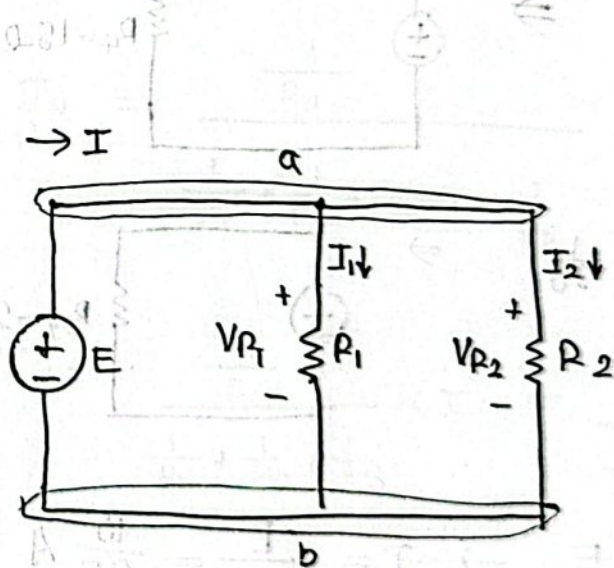
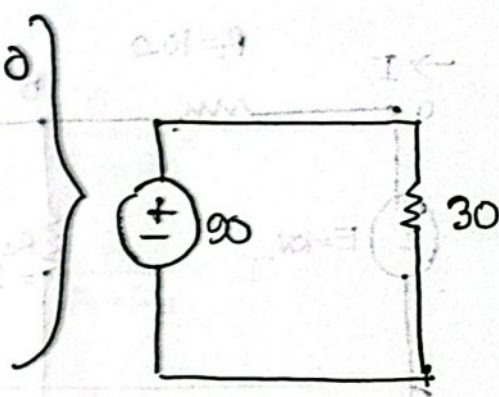
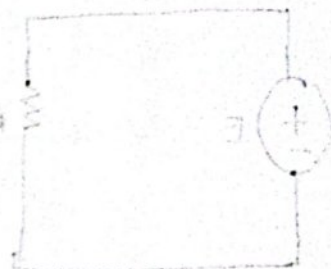


$$\Rightarrow R_{eq} = 30\Omega$$

$$\Rightarrow +80 - V_{R1} + 30 - V_{R2} - V_{R3} - 20 - V_{R4} = 0$$

$$\Rightarrow 90 - 10I - 7I - 5I - 8I = 0$$

$$\Rightarrow 90 - 30I = 0 \therefore I = 3A$$



$$E = V_{R1} = V_{R2} = V_{ab}$$

$$V_{ab} = V_{R1} = I_1 R_1$$

$$V_{ab} = V_{R2} = I_2 R_2$$

$$I \cdot E = I_1, O.C = I_1 + I_2$$

$$\therefore I = I_1 + I_2$$

* If Multiple elements are connected between two nodes, (a-b) then all the elements are connected in parallel.

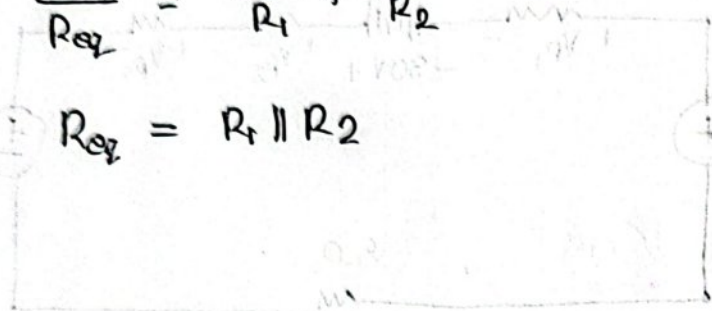
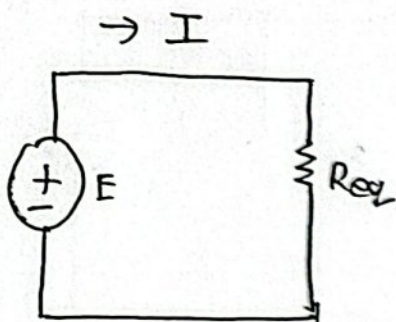
$$\therefore I = \frac{V_{R1}}{R_1} + \frac{V_{R2}}{R_2} \Rightarrow I = E \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$I = \frac{E}{R_{eq}}$$

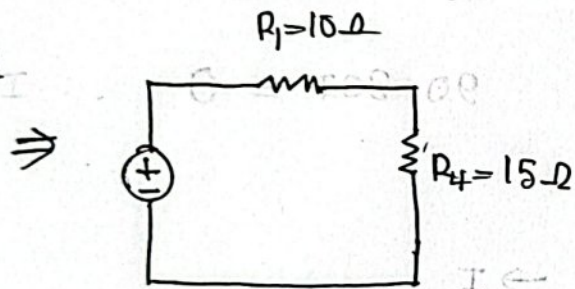
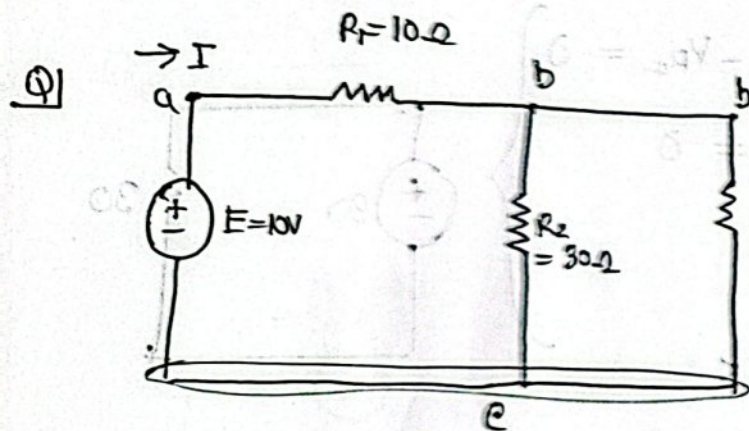
$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\therefore R_{eq} = R_1 \parallel R_2$$



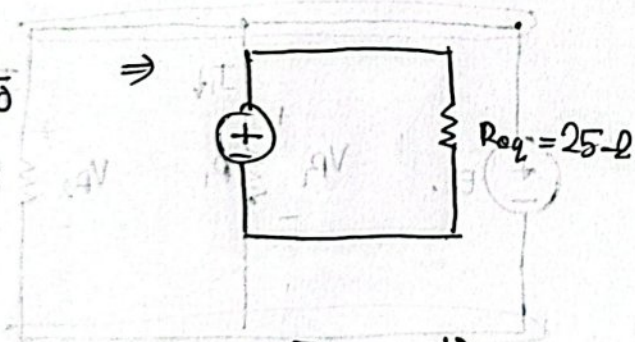
$$0.08 = \mu A$$



Soln: $R_4 = R_2 \parallel R_3$

$$\frac{1}{R_4} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{30} + \frac{1}{30}$$

$$\therefore R_4 = 15\Omega$$

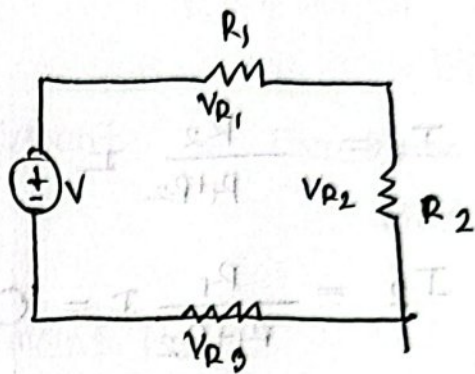


$$\therefore E - V_{R_{eq}} = 0 \Rightarrow IR_{eq} = E \Rightarrow I = \frac{E}{R_{eq}} = \frac{10}{25} A$$

$$\therefore R_{eq} = R_1 + R_2 \parallel R_3$$

$$\left(\frac{1}{10} + \frac{1}{15} \right) I = \frac{10V}{25} + \frac{10V}{15}$$

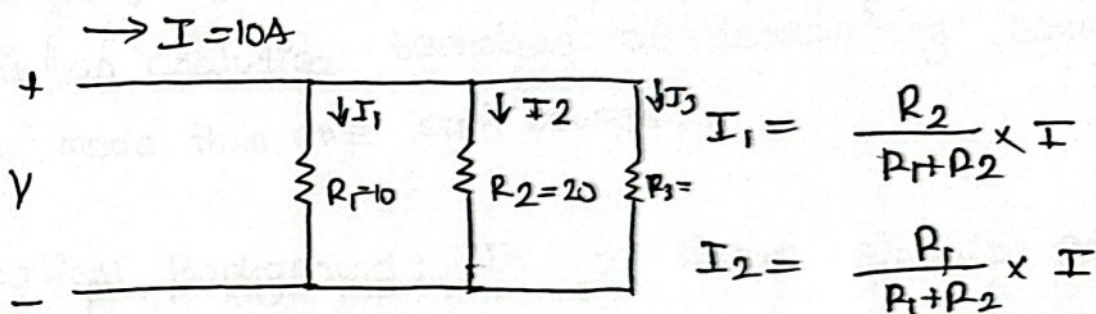
$$\frac{I}{R_{eq}} = I$$



$$V_{R1} = \frac{R_1}{R_1 + R_2 + R_3} \times V$$

$$V_{R2} = \frac{R_2}{R_1 + R_2 + R_3} \times V$$

$$V_{R3} = \frac{R_3}{R_1 + R_2 + R_3} \times V$$



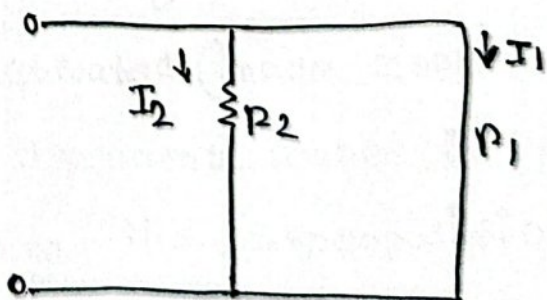
$$I_1 = \frac{R_2}{R_1 + R_2} \times I$$

$$I_2 = \frac{R_1}{R_1 + R_2} \times I$$

$$I_N = \frac{\frac{1}{R_N}}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}} \times I$$

Soln:

$$I_1 = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \times I = \frac{\frac{1}{10}}{\frac{1}{10} + \frac{1}{20} + \frac{1}{30}} \times 10$$

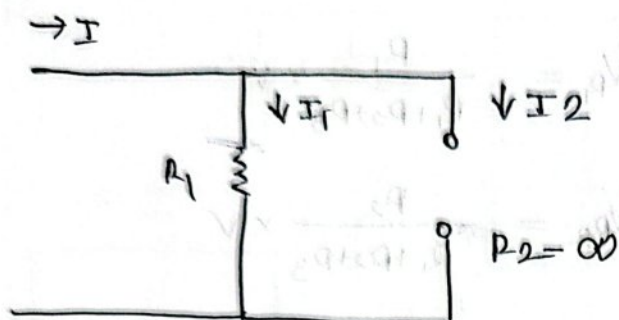


$$I_1 = \frac{R_2}{R_1 + R_2} \times I$$

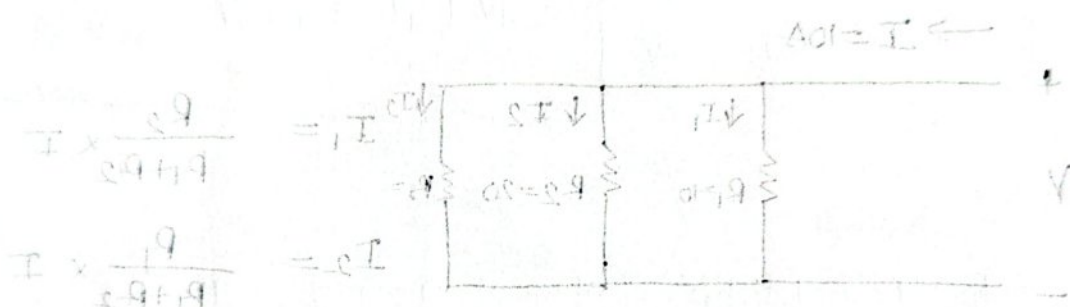
$$I_2 = \frac{R_1}{R_1 + R_2} \times I = 0$$

$$\therefore I = I_1 + I_2$$

$$I = I_1$$



$$\Rightarrow \begin{aligned} I_1 &= \frac{R_2}{R_1 + R_2} I \\ I_2 &= \frac{R_1}{R_1 + R_2} I = 0 \end{aligned}$$



$$I_n = I \times \frac{1}{nR}$$

$$\frac{1}{nR} = \frac{1}{R} + \frac{1}{R} + \dots + \frac{1}{R}$$

$$I \times \frac{1}{10} = I \times \frac{1}{R} \times \left(\frac{1}{10} + \frac{1}{20} + \frac{1}{30} \right)$$

$$I_1 = I \times \frac{R_2}{R_1 + R_2}$$

$$I_2 = I \times \frac{R_1}{R_1 + R_2}$$

$$I_1 + I_2 = I$$

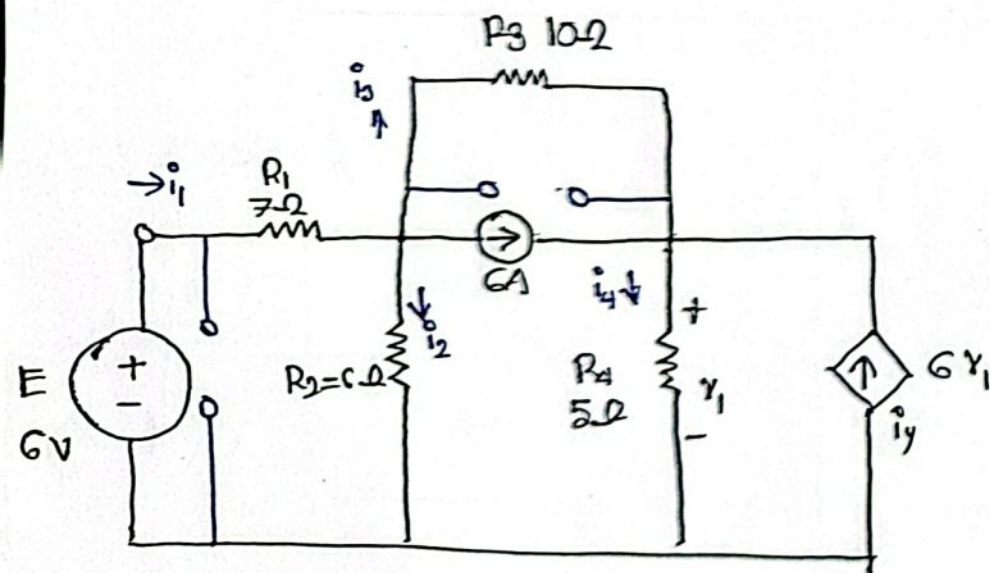


EEH01
Lee-09

- Remove the element through which the eqv Resistance which will be eliminated.
- Check if there's more than 1 sources.
- If independent source are there, remove the independent sources apart from the through which R_{eq} is measured.

Remove of independent source

- Current source: open
- Voltage source: short
- If there has independent source, don't remove dependent sources.
- Add a test source



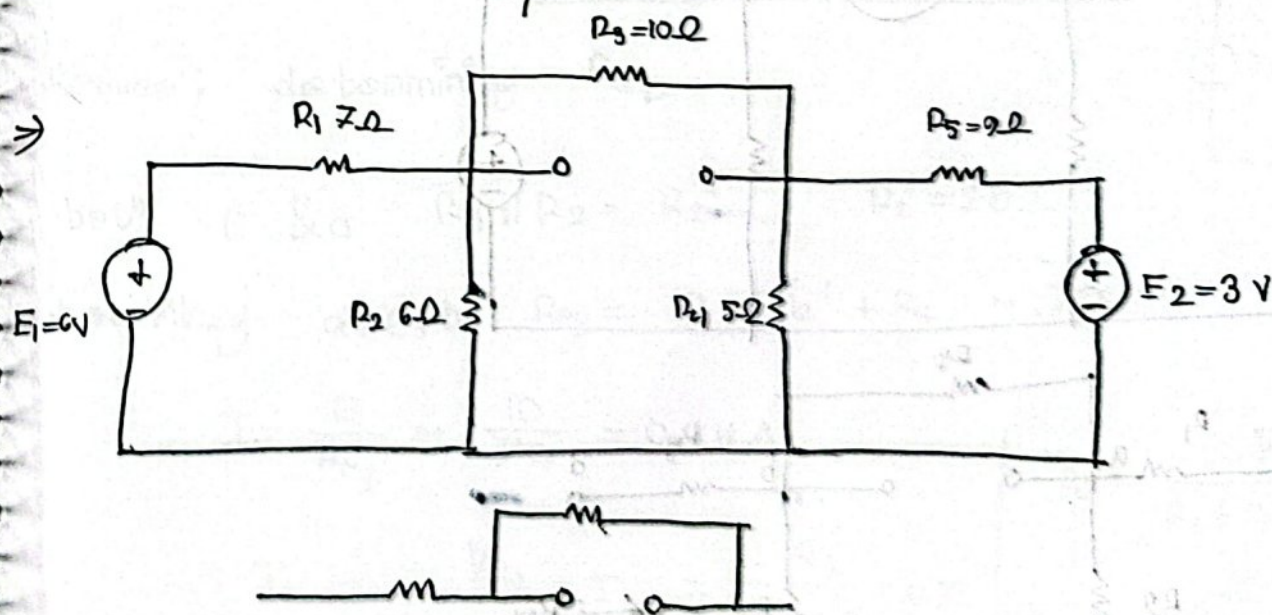
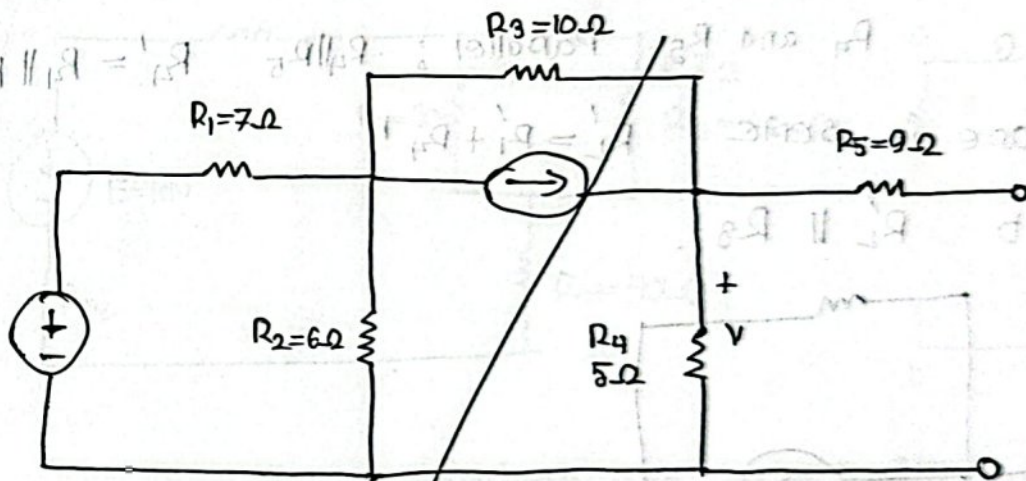
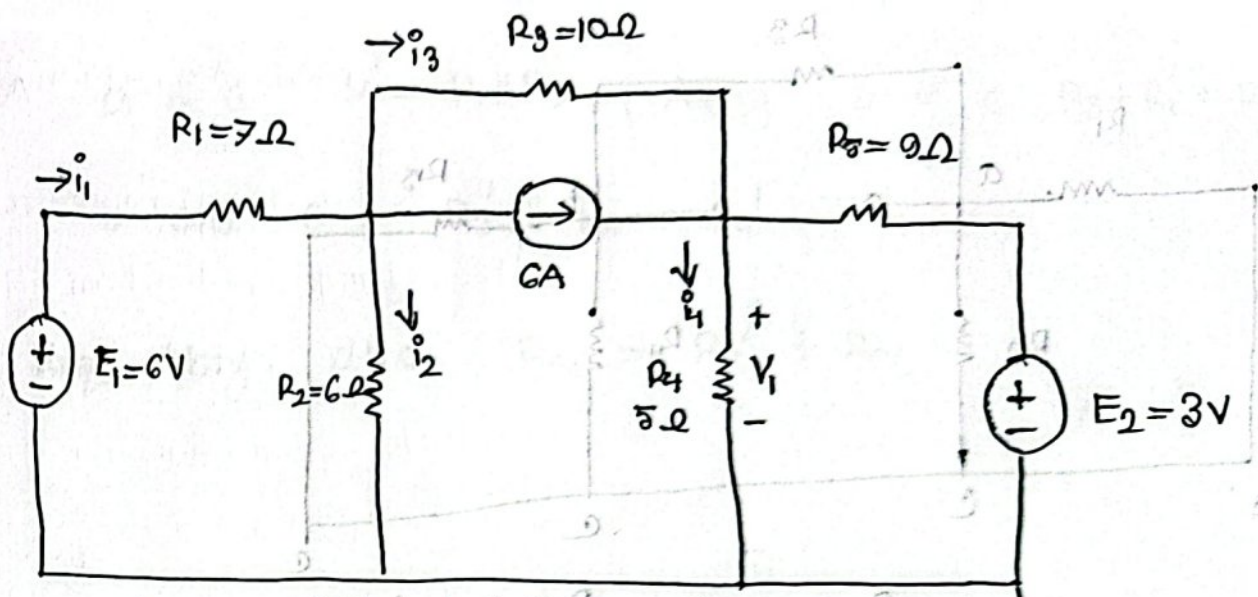
$$V_1 = I_4 R_4$$

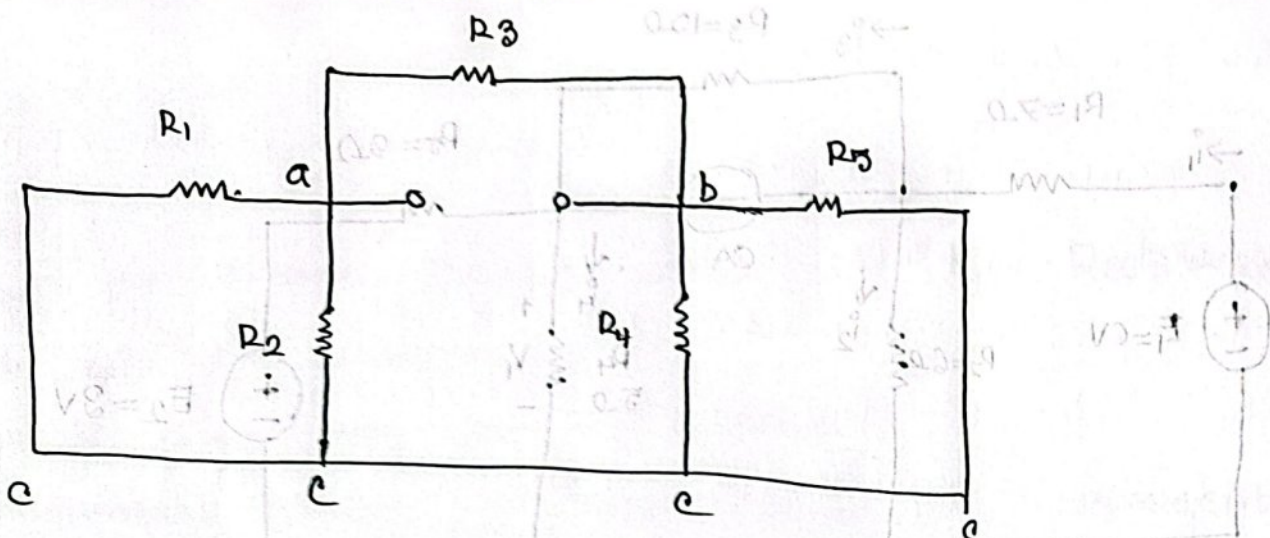
$$I_y = G \times I_4 \times R_4$$

① Open the current source

① Add a test source of $V_1 = 1V$

$$R_{eq} = \frac{V_x}{i_1}$$



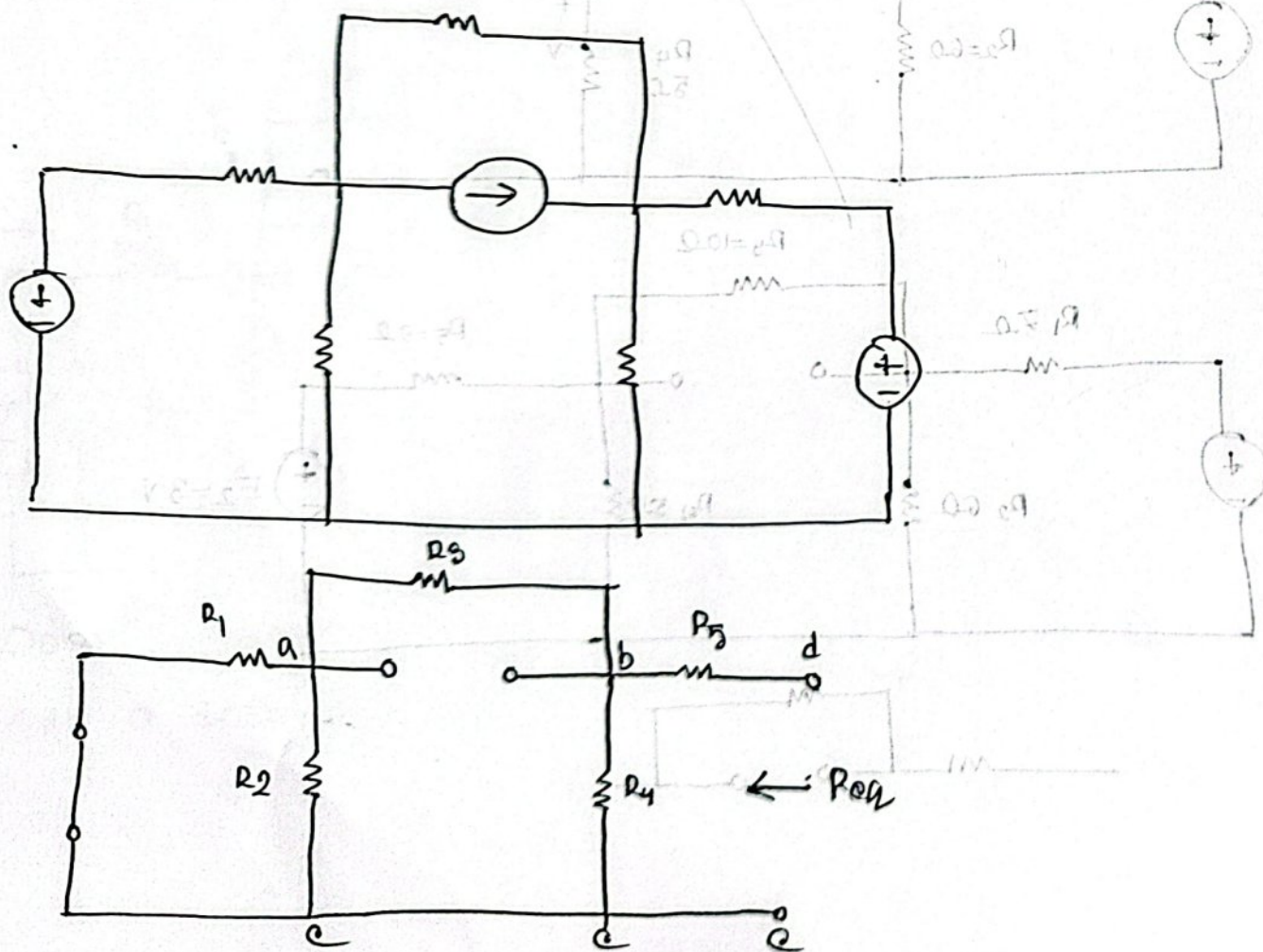


between a and c R_1 and R_2 Parallel : $R_1 \parallel R_2$ $R'_1 = R_1 \parallel R_2$

between b and c R_4 and R_5 Parallel : $R_4 \parallel R_5$ $R'_4 = R_4 \parallel R_5$

R'_1 and R'_4 are in series $R'_L = R'_1 + R'_4$

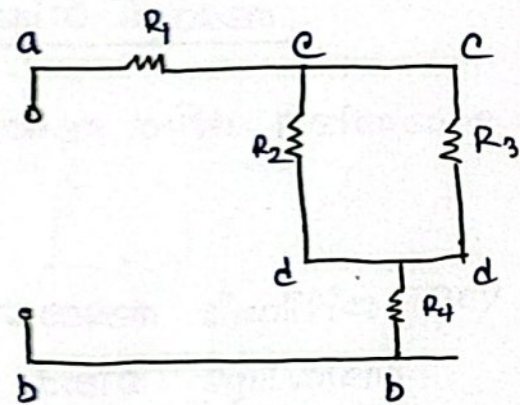
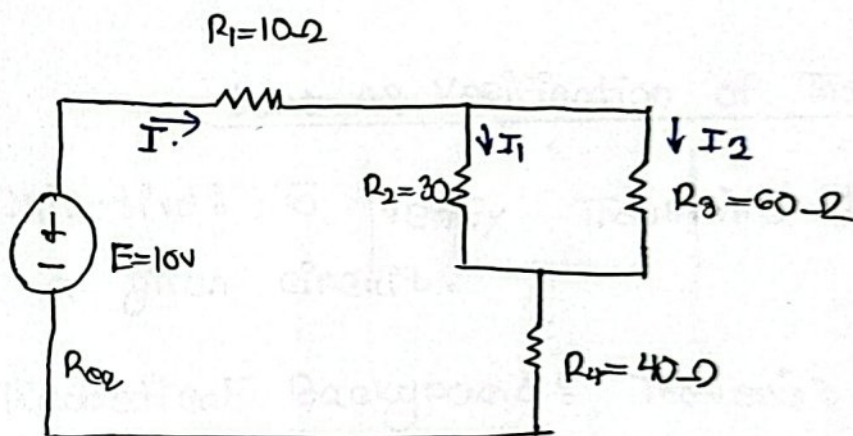
between a and b $R'_L \parallel R_3$



betn a & c $R' = R_1 \parallel R_2$ | Along b & c $R_3 + R' = R_3'$

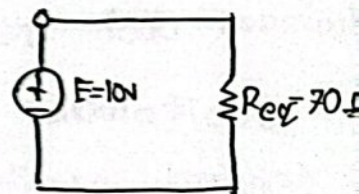
betn b and c: $R_3' \parallel R_4 = R_4'$

$\therefore$ Along dba $R_{eq} = R_4' + R_{eq}$



Reduce and Return Method

Reduce: determining R_{eq}



betn c & d $R_3 \parallel R_2 = R_2' \therefore R_2' = 20\Omega$

bet Along a, c, b $R_{eq} = R_1 + R_2' + R_4 = 70\Omega$

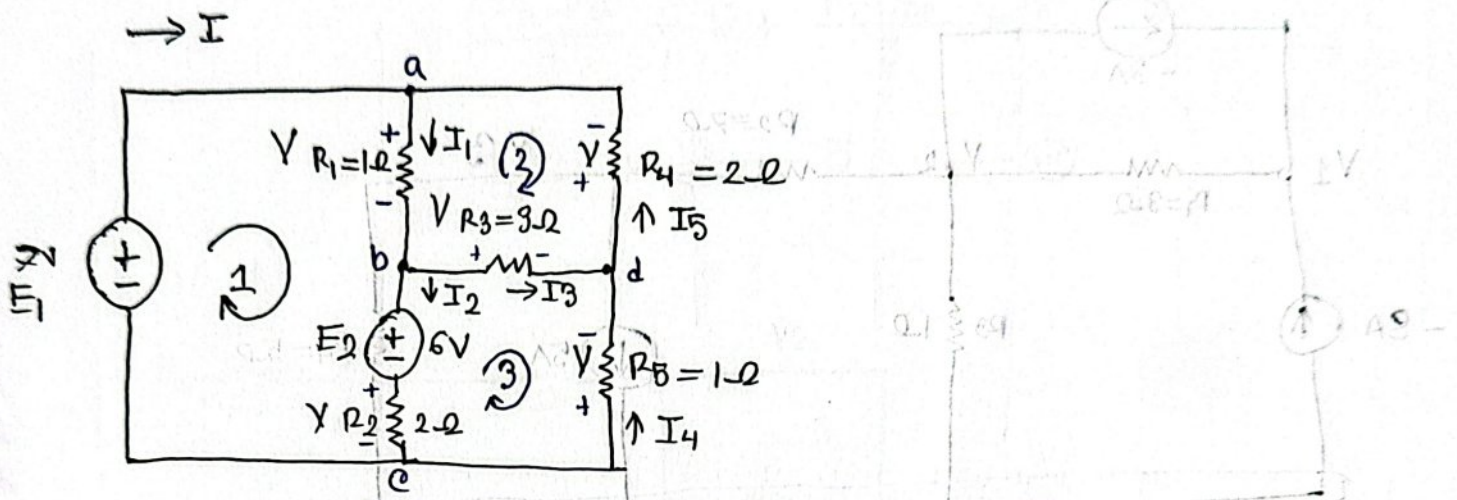
$$I = \frac{E}{R_{eq}} = \frac{10}{70} = 0.14A$$

$$\therefore I_1 = \frac{\frac{1}{R_2}}{\frac{1}{R_2} + \frac{1}{R_3}} I = \frac{R_3}{R_2 + R_3} \times I$$

$$I_2 = \frac{R_2}{R_2 + R_3} \times I$$

$$V_{R_4} = \frac{R_4}{R_1 + R_2' + R_4}$$

Quiz: Eq^v Resistance



6 Branches: ac, ab, bc, bd, cd, ad

Apply KVL in Independent loops

KVL @ 1: $+7 - VR_1 - 6 - VR_2 = 0$

$\Rightarrow 1 - 1I - 2I_2 = 0$ — (i)

KVL @ 2: $+VR_1 + VR_4 + VR_3 = 0$

$\Rightarrow 1I_1 + 2I_5 + 3I_3 = 0$ — (ii)

KVL @ 3:

$VR_2 + E_2 - VR_3 + VR_5 = 0$

$\Rightarrow 2I_2 - 3I_3 + 1I_1 = -6$ — (iii)

KCL @ a: $I_C = I + I_5$

$0.C = \neq 1$

$\therefore I + I_5 - I_1 = 0$ — (iv)

KCL @ b: $I_C = I_1$

$0.C = I_3 + I_2$

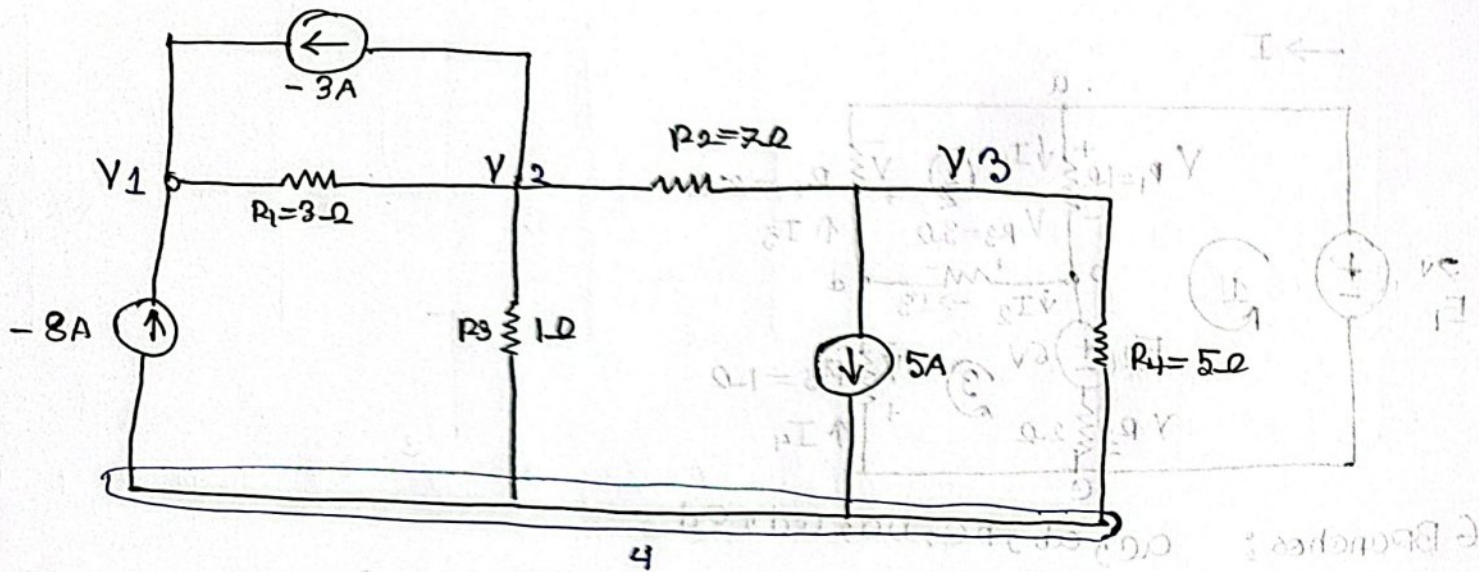
$\therefore I_1 - I_3 - I_2 = 0$ — (v)

KCL @ d: $I_C = I_3 + I_4$

$0.C = I_5$

$\therefore I_3 + I_4 - I_5 = 0$ — (vi)

Node Analysis :-



* No of Nodes = 4

⇒ Consider 1 Node as Ref Node.

No of Non Ref Nodes : 3

⇒ $V_{Ref} = 0V$

(All the current entering = -ve
At " " leaving = +ve)
(Always, unknown current is leaving the node working on)

KCL @ Node 1 :

$$-(-8) - (-3) + \frac{V_1 - V_2}{3} = 0$$

$$\Rightarrow V_1 - V_2 = -33 \quad \text{--- (i)}$$

KCL @ Node 2 :

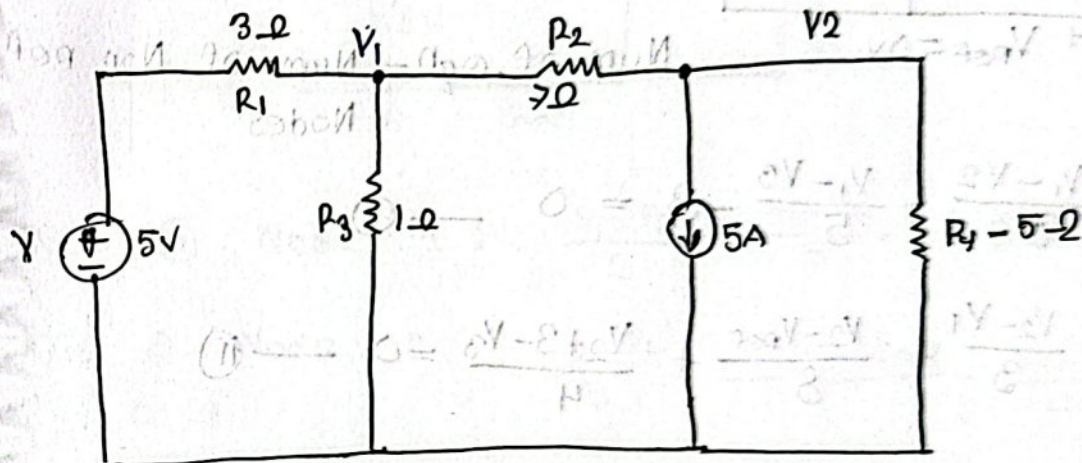
$$\frac{V_2 - V_1}{3} + \frac{V_2 - V_4}{1} + \frac{V_2 - V_3}{7} + (-3) = 0$$

$$\Rightarrow \frac{V_2 - V_1}{3} + \frac{V_2}{1} + \frac{V_2 - V_3}{7} = 3 \quad \text{--- (ii)}$$

KCL @ Node 3:

$$\frac{V_3 - V_2}{7} + 5 + \frac{V_3 - V_{ref}}{5} = 0$$

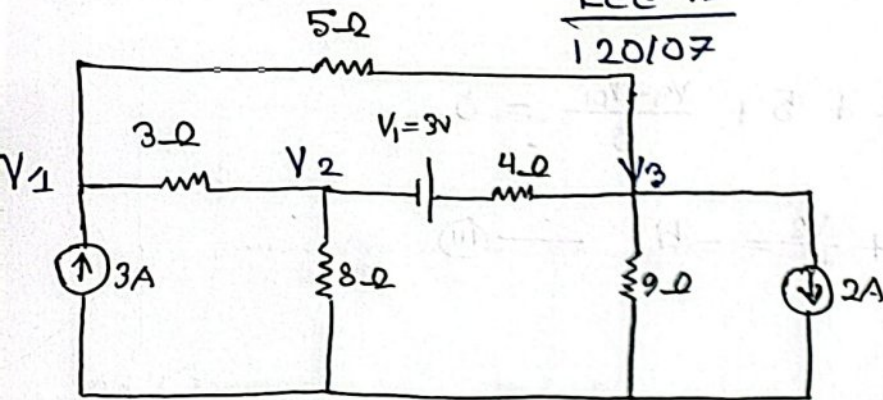
$$\Rightarrow \frac{V_3 - V_2}{7} + \frac{V_3}{5} = -5 \quad \text{--- (iii)}$$



KCL @ Node 1: $\frac{V_1 - 5}{R_1} + \frac{V_1 - V_{ref}}{R_3} + \frac{V_1 - V_2}{R_2} = 0 \quad \text{--- (i)}$

KCL @ Node 2: $\frac{V_2 - V_1}{7} + \frac{V_2 - V_{ref}}{5} + 5 = 0 \quad \text{--- (ii)}$

LeC-12
120/07



$I_3 = ?$ (current through 3Ω)

Nodes = 4

Node 4: Ref Node

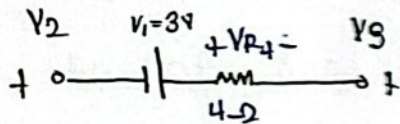
$V_{ref} = 0V$

Num of eqn = Num of Non ref Nodes

$V_1 - V_{ref} = V_1$

KCL @ Node 1: $\frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{5} - 3 = 0$ — (i)

KCL @ Node 2: $\frac{V_2 - V_1}{3} + \frac{V_2 - V_{ref}}{8} + \frac{V_2 + 3 - V_3}{4} = 0$ — (ii)



KCL @ Node 3:

$\frac{V_3 - 3 - V_2}{4} + \frac{V_3 - V_{ref}}{9} + \frac{V_3 - V_1}{5} + 2 = 0$ — (iii)

$\Rightarrow +V_2 + 3 - V_{R4} - V_3 = 0$

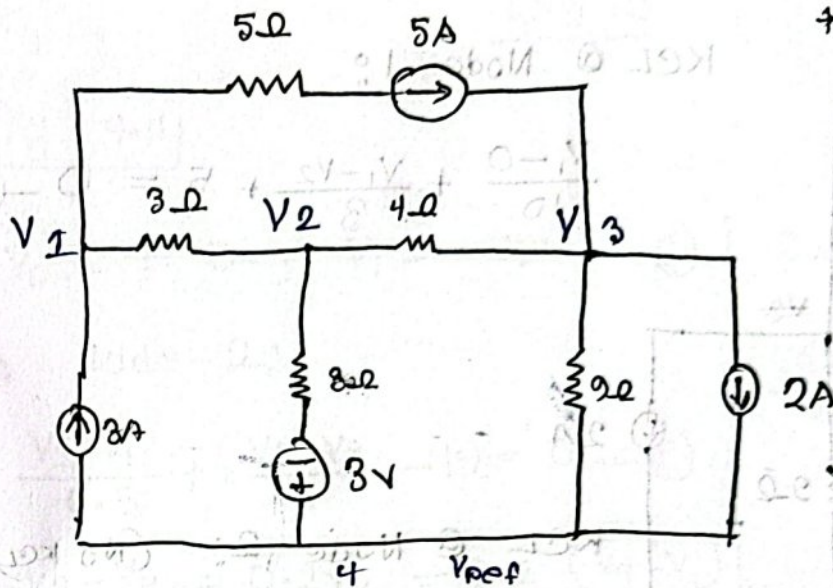
$\Rightarrow V_2 + 3 - 4I_4 - V_3 = 0$

$\Rightarrow I_4 = \frac{V_2 + 3 - V_3}{4}$

$I_4' = \frac{V_3 - 3 - V_2}{4}$

* if current is already known, don't need to do

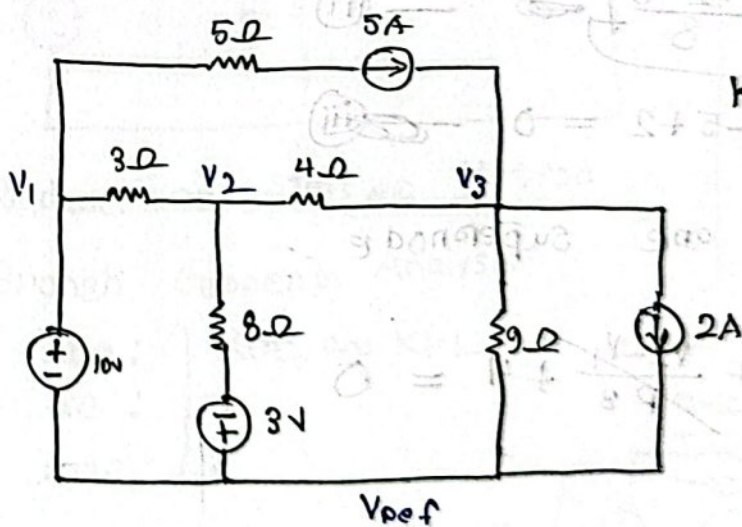
$$\frac{V_1 - V_3}{5} !$$



KCL @ Node 1: $\frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{5} - 5 = 0$ — (i)

KCL @ Node 2: $\frac{V_2 - V_1}{3} + \frac{V_2 + 3 - V_{ref}}{8} + \frac{V_2 - V_3}{4} = 0$ — (ii)

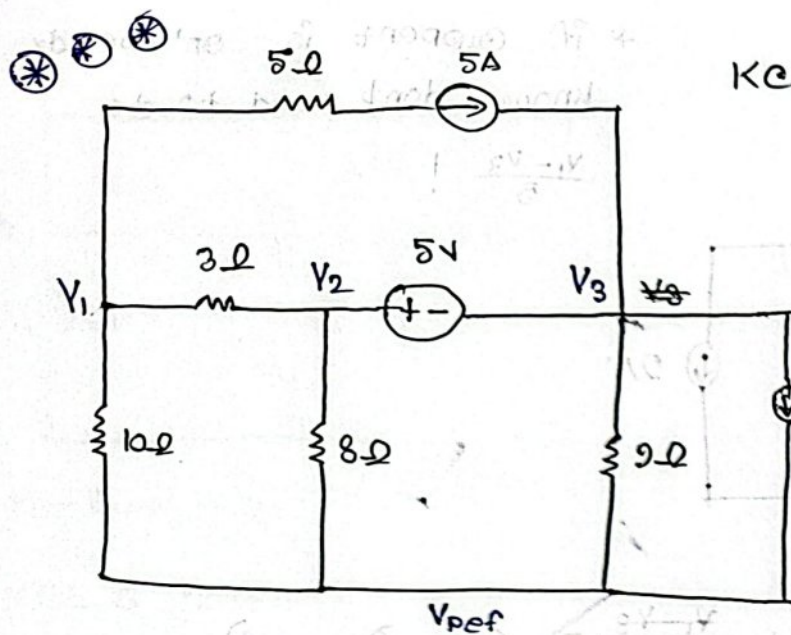
KCL @ Node 3: $\frac{V_3 - V_2}{4} + \frac{V_3 - V_{ref}}{9} + 2 - 5 = 0$ — (iii)



KCL @ Node 1:

$$\frac{V_1 - V_3}{5} = V_1 - V_{ref} = 10$$

$\therefore V_1 = 10$ — (i)



KCL @ Node 1:

$$\frac{V_1 - 0}{10} + \frac{V_1 - V_2}{3} + 5 = 0 \quad \text{--- (i)}$$

KCL @ Node 2: (No KCL)

$$V_2 - V_3 = 5 \quad \text{--- (ii)}$$

KCL @ Node 3: (No KCL)

$$V_3 = 0$$

SuperNode (2 & 3): Replace the Voltage source Betⁿ Node 2 & 3 with short, ckt.

~~KCL @ Node 2:~~ $\frac{V_2 - V_1}{3} + \frac{V_2 - 0}{8} + \dots = 0$ --- (iii)

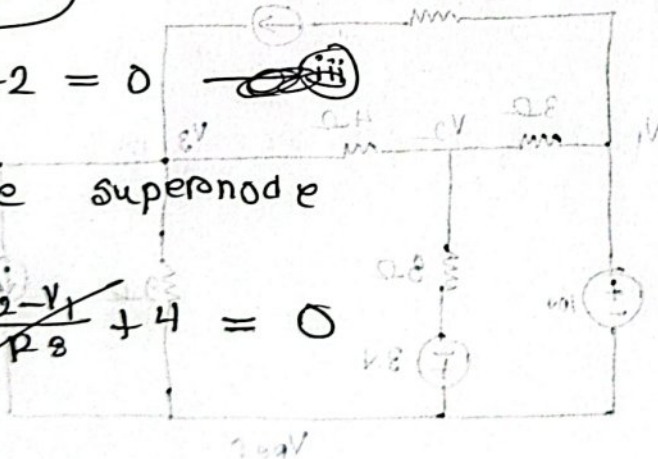
~~KCL @ Node 3:~~ $\frac{V_3 - 0}{9} + \dots = 0$ --- (iv)

Fig 8.63: Node 1 & 2 are supernode

$$\frac{V_1}{R_1} + \frac{V_1 - V_2}{R_3} - 6 + \frac{V_2}{R_2} + \frac{V_2 - V_1}{R_3} + 4 = 0$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} = 2 \quad \text{--- (v)}$$

$$V_1 - V_2 = 12V \quad \text{--- (vi)}$$



Loc 13

Branch Current Analysis
Nodal
Power Measurement
Equivalent Resistors

Fig 4.11

$$V_{ref} = 0V, V_1 = -12V \text{ --- (i)}$$

@ Node 2:

$$\frac{V_2 - V_1}{0.5} + \frac{V_2 - V_3}{2} - 1 = 0 \text{ --- (ii)}$$

$$V_3 - V_4 = 0.2V_4 = 0.2(V_4 - V_1) \text{ --- (iii)}$$

Supernode eqn betn 3 & 4

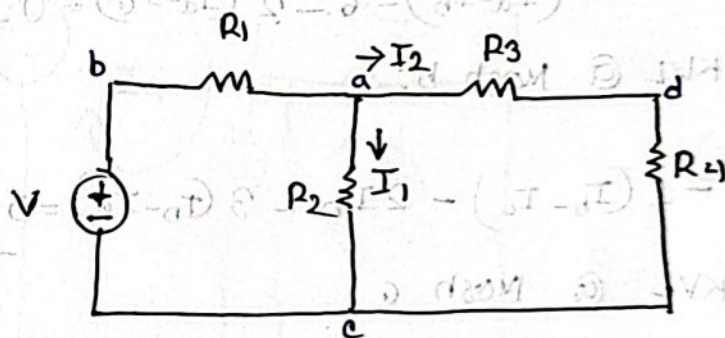
$$\left(\frac{V_3 - V_1}{2} - 0.5(V_2 - V_1) \right) + \left(\frac{V_4 - V_1}{2.5} + \frac{V_4 - V_{ref}}{1} \right) = 0$$

KCL @ N3

KCL @ Node 4

Topic: Mesh Analysis

Ohm's law, KVL in single loop



Reduce and Return Method

Branch Current Analysis:

- ⇒ abc: KCL and KVL
- ⇒ ac:
- ⇒ adc:

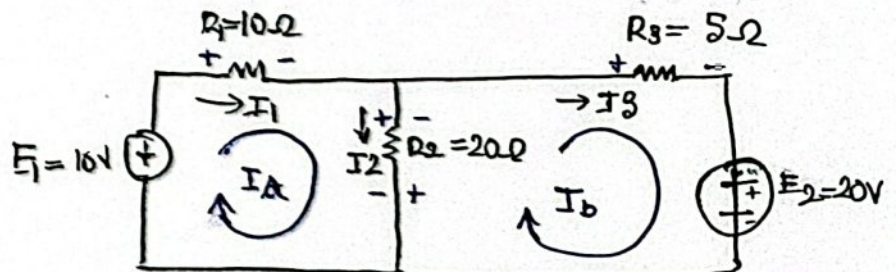
⇒ Mesh = Independent loop

$$R_{eq} = R_1 + (R_2 || (R_3 + R_4))$$

$$I = \frac{V}{R_{eq}}$$

$$I_1 = \frac{R_3 + R_4}{R_2 + R_3 + R_4} \times I$$

$$I_2 = \frac{R_2}{R_2 + R_3 + R_4} \times I$$



Step 1: Find out Independent loops/mesh

Step 2: For each mesh assign a current & Any direction

Step 3: KVL @ Mesh 1

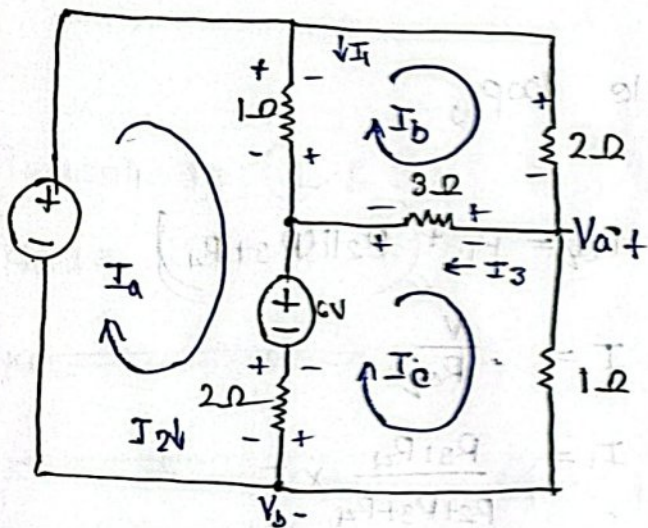
KVL @ Mesh 1: $+E_1 - I_a R_1 - (I_a - I_b) R_2 = 0$ — (i)

$V_{R_2} = R_2 (I_b - I_a)$
 $V_{R_2} = (I_a - I_b) R_2$

* Consider the mesh current that you are working on is the higher current among all the mesh currents! (i)

KVL @ Mesh 2: $-R_2 (I_b - I_a) - I_b R_3 - E_2 = 0$ — (ii)

$(V - 1V) \cdot 0.0 = V \cdot 0.0 = 1V - 0V$
 (iii) $\leftarrow$



KVL @ Mesh a:

$+7 - 1(I_a - I_b) - 6 - 2(I_a - I_c) = 0$ — (i)

KVL @ Mesh b:

$-1(I_b - I_a) - 2I_b - 3(I_b - I_c) = 0$ — (ii)

KVL @ Mesh c:

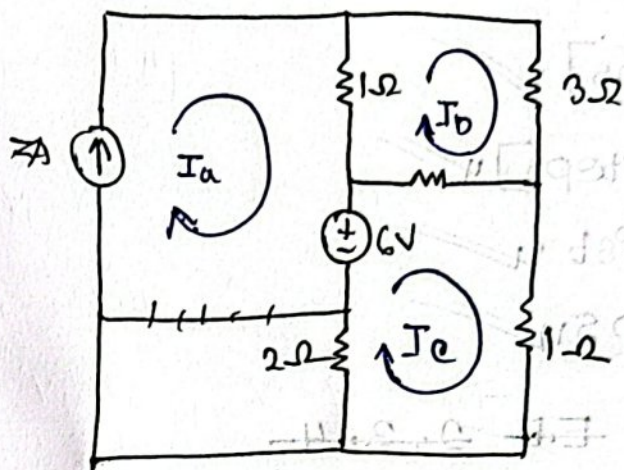
$-2(I_c - I_a) + 6 - 3(I_c - I_b) - 1I_c = 0$ — (iii)

$I_1 = I_a - I_b$
 $I_3 = I_b - I_c$
 $I_2 = I_a - I_c$
 $+2I_2 + 6 - 3I_3 - V_{ab} = 0$



Mesh = independent loop

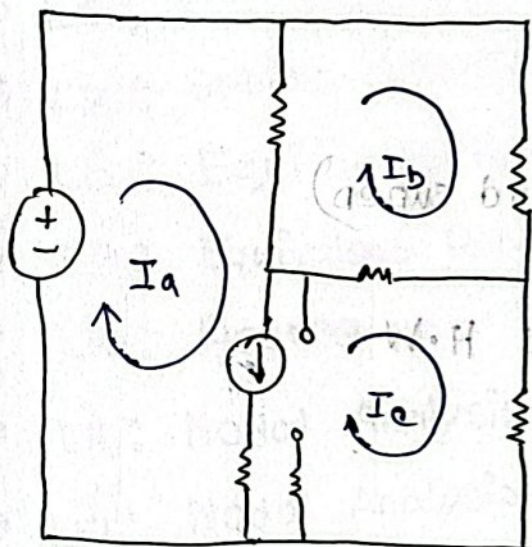
Step 1: Find independent loops (meshes)
 Step 2: For each mesh, assign a clockwise current
 Step 3: KVL @ Mesh 1



~~KVL @~~ $I_a = 7A$ — (i)

$-1(I_b - 7) - 2I_b - 3(I_b - I_c) = 0$ — (ii)

$-2(I_c - 7) + 6 - 3(I_c - I_b) - 1I_c = 0$ — (iii)



KVL @ Mesh a:

~~$+6 - 1(I_a - I_c) = 0$~~ $I_a - I_c = 7$ — (i)

KVL @ mesh b:

$-1(I_b - I_a) - 3I_b - 3(I_b - I_c) = 0$ — (ii)

Super Mesh!

$\Rightarrow \underbrace{(+6 - 1(I_a - I_c))}_{\text{Mesh a}} + (3(I_c - I_b) - 1I_b) = 0$